

GENERAL THETA FOUNDATIONS I: CONTINUATION COMPLEXITY AND STABLE CAUSAL CERTIFICATION

QIAN QI

ABSTRACT. We study marked causal experiments with separately charged simulator and verifier memory. Compatible continuation responses characterize realizability at a prescribed width profile. For Boolean streaming tests we turn a deficit of continuation states into a quantitative statistical loss, and determine exactly the widths attaining the optimal odd-sample binary reset value. On a genuinely interacting two-sphere family, the private continuation restriction yields one rational audit with 400 training trials and a nineteen-bit retained register whose expected discrepancy exceeds $2/5$. The same audit remains valid while contact distance changes the collision time and microscopic path. A common-preparation Gaussian comparison transports likelihoods and posterior processes; a bounded terminal tilt then gives global, rather than localized, convergence of the associated entropic backward integrands and their martingale integrals. The certificate and this conditional-law consumer are consequences of one microscopic transport estimate. Hidden and visible selectors retain distinct costs, and removing the collision makes the finite resource gaps vanish.

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1. INTRODUCTION

A causal reduction consumes observations, emits reports and retains a state. The dimension of that state and the visibility of persistent randomization are part of the mathematical problem. The same is true of a test intended to distinguish a reduction from its target: a test given the complete candidate law has a different information budget from a test which reads finitely many trials through a finite register. We study these two sides of causal comparison together, without convexifying an unpriced private resource class.

The organizing object is a family of normalized continuation responses, with one compatible list at each cut. Compatibility means that every one-step input–output shift is a positive combination of the next list. It characterizes realizability at a prescribed register profile. For zero–one terminal decisions, the lists reduce to residual equivalence classes. This allows an exact calculation of the memory needed to attain a statistical testing value, as opposed to an upper bound for the size of an unconstrained certificate. We then use a private continuation restriction to construct a small streaming certificate for an interacting microscopic experiment. The same certificate and its conditional-law consumers are transported through a change in the microscopic trajectory.

1.1. Main results. The resource signature distinguishes fresh private randomness from an independent persistent hidden selector and from a visible selector. An actual terminal mark belongs to the experiment; the online simulator cannot read it, but the terminal comparison can. A reset auditor is a separate machine. It reads specified independent completed trials of a candidate held fixed during the audit. Its schedule is known and its retained register is charged. It has no access to the candidate law or to the candidate’s private state.

Theorem 1.1 (Continuation complexity and stable causal certification). *With these information conventions the following statements hold.*

(i) A finite causal kernel is realizable at a width profile if and only if it admits a compatible normalized continuation cover of that profile. For a Boolean terminal test, the exact minimal profile is the number of distinct continuation responses at each cut, even when the implementing machine is stochastic. If a cut has $r \geq 2$ distinct responses and only K retained states, its total wordwise error is at least $(r - K)_+/(r - 1)$.

(ii) For $0 < h < 1/2$, consider $n = 2m + 1$ independent binary samples from one of $\text{Ber}(1/2 - h)$ and $\text{Ber}(1/2 + h)$, and the signed event-discrepancy audit against $\text{Ber}(1/2)$. Its unrestricted value is $h\|P_+ - P_-\|_{\text{TV}}$. The exact training widths needed to attain it are

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n.$$

A smaller width at any cut has the strictly positive value deficit (5.4). Thus the continuation invariant controls both an implementation and a statistical obstruction.

(iii) In the prepared two-sphere experiment of Theorem 14.2, one rational audit using 400 candidate training resets, two validation trials, and at most 504008 retained auditor states has expected discrepancy greater than $2/5$ against every private width-one candidate. The same audit works for every contact diameter $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. Removing the collision, by taking $d \leq 2/5$, makes all three finite-experiment deficiencies zero.

(iv) With positive noise fixed, changing d changes both the collision time and the microscopic path. If $\rho = |d' - d| + 2700|d' - d|^2$, the same audit's expected score changes by at most $\min\{1, \sqrt{\rho}/(2\sigma)\}$. Along $d' \rightarrow d$, the microscopic paths, full and projected likelihoods, and bounded posterior processes converge jointly in the senses specified below. For the associated bounded entropic backward equation, values converge in supremum L^2 and integrands converge globally in time-integrated L^2 , including on the common-latent coupling. Innovations converge uniformly in probability, and backward martingale integrals converge in supremum L^2 .

Part (i) follows from Theorems 3.2 and 5.1; part (ii) is Theorem 5.2. Parts (iii)–(iv) are the single composition theorem proved in Section 14, using the global transport theorem of Section 12. The two registers in part (iii) have different roles: the candidate remains private width one, while the auditor's nineteen-bit implementation has its own explicit budget. This is not a statement that the candidate can borrow the auditor's memory.

The finite-data construction is materially tied to the resource geometry. The private width-one candidate has independent reports and a fair independent mark. Only two report-pair probabilities determine the four relevant marked coordinates. Two truncated empirical counts therefore replace a full multinomial table. A direct regret inequality gives the 400-sample rule, and its deterministic continuation states give the register bound. The collision supplies the separating marked target. The common-preparation variation estimate then transports the very same rule, not a reoptimized rule, together with the global backward representation.

1.2. The general comparison framework. For a compact marked behavior image B and fixed target rows Q_j , set

$$\delta(B, Q) = \min_{b \in B} \max_{j, A} \{Q_j(A) - b_j(A)\}.$$

Complements supply both signs. Theorem 2.2 supplies compactness, exact factorization, priced composition and perturbation at the declared signature. Theorem 4.1 gives an attained finite-reset moment dual on the original, possibly nonconvex B and $0 \leq \delta - v_n^{\text{aud}} \leq \sqrt{D/n}$. Its dual chooses one candidate for the entire replicated experiment; its barycenter cannot replace that candidate mixture. Theorem 4.3 proves a matching uniform sampling exponent. The new memory obstruction is different: it identifies exactly when a streaming auditor attains the unconstrained value, and gives a strict deficit otherwise. We do not exchange minimax on the nonconvex set of fixed-width auditors.

The revealed-behavior representation and its intrinsic Bernstein hierarchy remain separate tools, with their full proofs and exact nonlinear example. Their evaluator sees an exact law, unlike the reset auditor. The formal rational chart algorithm and physical probability enclosures retain this distinction. Effective here means an exact terminating construction under specified rational inputs and positive-noise hypotheses, not polynomial bit complexity.

1.3. Classical inputs and the contribution. Statistical comparison and randomization have a classical foundation in Blackwell and Le Cam [1, 2, 13]; filtered comparison predates this work [3, 19]. We do not claim priority for introducing a filtration or a randomization criterion. The original proof-level crosswalk with Norberg remains unverified; no assertion about absence of the present results from that work is made.

Stable positive-cone realization is classical [23], as is residual-state minimization [16]. The continuation theorem is a finite controlled normalization with a simultaneous width profile. The zero-one residual characterization is proved here in a stochastic implementation model, not claimed as a new deterministic automata theorem. The added step is the overlap inequality (5.2), which converts insufficient continuation width into statistical regret and an exact optimal-value threshold for the binary reset game. The quantitative penalty is explicit, but not claimed to be a sharp joint asymptotic law in sample size, memory and error.

The minimax theorem, moment support reduction, empirical second moments, two-point testing, and polynomial positivity are classical tools [18, 20, 21, 22]. Likewise the scattering formula, Bayes' formula, Girsanov's theorem and martingale inequalities are inputs, not discoveries. The global backward estimate uses a denominator-free integrated posterior comparison and a bounded terminal tilt. It improves the localized transport estimate while retaining the same actual terminal condition and driver. The contribution offered here is the information-budgeted theorem chain: compatible continuations, constrained testing value, an executable microscopic certificate, and global transport of its conditional-law consumer.

1.4. Microscopic and programmatic scope. The physical family has two labeled spheres and one nongrazing collision. The correlation between the impact variable and the actual mark is prepared; the collision creates an observable transverse signal of that variable. The feedback is an acquisition gate, not a mechanical force. Varying contact diameter is a genuinely changing microscopic path, not a Boltzmann-Grad limit. The global transport theorem assumes a common preparation, bounded signals and fixed positive noise. Its bounded scalar entropic equation has an observation-measurable conditional certainty equivalent at terminal time, not an unobserved latent terminal payoff.

This gives an explicit conditional-law consumer in the larger theta-theory program, including changing paths, likelihoods, projected stochastic exponentials and the complete bounded backward representation. It does not manufacture a dependence of the independent A2 geometric chain, nor use an unverified kinetic resolvent or rigidity theorem as an input. The wider program and its historical targets are retained in full. Their source assertions, current proof credit and independent-review status are recorded separately in the accompanying statement ledger.

The canonical dependency diagram is in Appendix A. Every mathematical module used here has a full proof in this article. The complete predecessor development is preserved as a companion; its historical introductions are not substitutes for the present theorem statements.

2. INTRINSIC CAUSAL DEFICIENCY

2.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments: for each parameter and control history they give the joint law of the report path and the actual

mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m &\in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' &\in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \quad (2.1)$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \quad (2.2)$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta,g}^{q^u} - Q_{\theta,g} \right\|_{\text{TV}}, \quad (2.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (2.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed-marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 2.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

2.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta,g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta,g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (2.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta,g}(B) - P_{\theta,g}^d(B). \quad (2.5)$$

Including both an event and its complement includes both signs.

Theorem 2.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta,g}(B) - P_{\theta,g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (2.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (2.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (2.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (2.8)$$

(iii) *Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then*

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (2.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (2.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (2.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) *For compatible experiments E, F, G and profiles K, L ,*

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (2.12)$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multi-affine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (2.2); the same proof includes λ and the r policies in (2.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (2.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (2.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv}\{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (2.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (2.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (2.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (2.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (2.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values. For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (2.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product

with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 2.3 (What the converse says). The dual (2.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

2.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 2.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (2.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (2.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/(1 - \varepsilon)^{-1} \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1 - \varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (2.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (2.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (2.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and $A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 2.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

2.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (2.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 2.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 2.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 2.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual

terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 2.6 (Finite computation and general carriers). For rational finite instruments, (2.6) and the fixed- r formulations are finite polynomial feasibility problems; (2.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

3. CONTINUATION GEOMETRY AND AN EXACT MEMORY OBSTRUCTION

3.1. Normalized continuation laws. We first separate the transducer from the source to which it is attached. There are L epochs with finite input alphabets X_t and output alphabets Z_t . A causal kernel $p(z_{1:L} \mid x_{1:L})$ is nonnegative, normalized for every input word, and has output-prefix marginals independent of later inputs. The current input can include both a source report and the current control. It never includes an unavailable parameter, a terminal mark, or an earlier output not retained by the machine.

Let $\mathcal{C}_t$, $0 \leq t \leq L$, be the finite-dimensional polytope of normalized causal kernels from $x_{t+1:L}$ to $z_{t+1:L}$; put $\mathcal{C}_L = \{1\}$. For $v \in \mathcal{C}_t$ define the shift

$$(S_{x,z}v)(z_{t+2:L} \mid x_{t+2:L}) = v(z, z_{t+2:L} \mid x, x_{t+2:L}).$$

Its total mass $v(z \mid x)$ is independent of the future input word. A zero shift is allowed.

Definition 3.1 (Compatible continuation polytopes). A continuation cover of width $K = (K_0, \dots, K_L)$ consists of lists $V_t = (v_{t,m})_{m \leq K_t}$ in $\mathcal{C}_t$ such that, for every m, x, z ,

$$S_{x,z}v_{t,m} \in \text{cone } V_{t+1}. \quad (3.1)$$

It covers p when $p \in \text{conv } V_0$. Lists may repeat generators and may contain unreachable generators. Thus unused states can be adjoined.

Theorem 3.2 (Continuation realization with a width profile). *A causal kernel p has a stochastic transducer realization with an initial K_0 -valued register and K_t retained states after epoch t if and only if it has a compatible continuation cover of width K . Initial randomization is part of the charged initial register. With a deterministic initial state the condition is $p = v_{0,1}$ and $K_0 = 1$.*

Consequently the set of feasible width profiles depends only on the causal kernel and its declared input interface, not on a row presentation or a labeling of the hidden states. Every matrix of conditional continuation probabilities at cut t has nonnegative rank at most K_t . Individual rank bounds do not replace the simultaneous shift conditions (3.1).

Proof. Given a transducer, let $v_{t,m}$ be the suffix response when its register at time t equals m . The remaining rows define this response even for an unreachable state. If the next row is $q_{t+1}(z, m' \mid m, x)$, conditioning on its output and new state gives

$$S_{x,z}v_{t,m} = \sum_{m'} q_{t+1}(z, m' \mid m, x) v_{t+1,m'}.$$

The initial state distribution expresses p as a convex combination of V_0 .

Conversely choose nonnegative coefficients in every cone membership in (3.1). There are only finitely many choices to make. Since every generator on the right has mass one for each future

input word, the sum of the coefficients for a fixed m, x, z is $v_{t,m}(z \mid x)$. Summing over z gives one. The coefficients are therefore stochastic rows $q_{t+1}(z, m' \mid m, x)$. Choose an initial distribution representing p in $\text{conv } V_0$. Backward induction from $\mathcal{C}_L = \{1\}$ proves that the constructed machine has exactly the response $v_{t,m}$ from each register state. In particular its initial response is p . Equality for open-loop input words implies equality under adaptive input selection: at each node of an input–output history tree, the same causal conditional probability is used on the two sides.

For the rank assertion, condition on any positive-probability observed history at the cut. Its continuation response is a convex combination of $v_{t,m}$ with weights equal to the posterior register distribution. Stack these responses as rows, and use all future input–output pairs as columns. This is a nonnegative factorization through K_t generators. The converse construction required the same generator families at adjacent cuts and all their shifts; an unrelated factorization of each cut matrix provides no such compatibility. The characterization, being stated entirely in terms of these response polytopes, also proves the asserted invariance. $\square$

Attach such a causal kernel to a marked source experiment. The output law is obtained by contraction with the source’s causal instruments, while its actual mark is retained. Hence zero private deficiency is equivalently the existence of a compatible continuation cover whose contraction equals the target marked behavior in every parameter and feedback row. This is the geometric form of the factorization in Theorem 2.2; no conditional dependence on the mark is introduced.

A hidden selector chooses one such cover independently of the source. It is a convex combination of complete realizations, not a convexification inside every retained-state simplex at the original width. Its private implementation uses the disjoint union of the covers and keeps the selector in the state. A visible selector keeps those fibers in the compared law. The nonnegative fiber argument in Theorem 2.2 therefore continues to give equality of the visible and private zero sets.

3.2. An erasure cut. A simple family makes the width obstruction exact, including its error. Let X take values in $[N]$ with law π . Before a specified cut the source reveals X . After the cut it has no reports correlated with X . The controller has no feedback channel carrying X across the cut, and all earlier emitted reports are prescribed constants. The only permitted cross-cut information of the simulator is its K -valued register and any independent selector explicitly included in its signature. The terminal mark is the actual $W = X$. The target’s terminal report is X . All memory at other checkpoints is unrestricted.

Theorem 3.3 (Exact cut spectrum). *For this experiment, for private, hidden and visible signatures alike,*

$$\delta_K = 1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}. \quad (3.2)$$

For uniform X , $\delta_K = (1 - K/N)_+$. Thus an erasure cut for m unbiased bits requires $K \geq (1 - \epsilon)2^m$ whenever its error is at most ϵ . The bound holds for every surrounding width profile; equality is attained when the other checkpoints are unrestricted. If every $\pi_x > 0$, exact transmission has nonnegative cut rank N .

Proof. With all post-cut randomness averaged into the decoder, any private machine gives kernels $e(m \mid x)$ and $d(z \mid m)$. Its joint terminal law is $\pi_x \sum_m e(m \mid x) d(z \mid m)$. Since the target law is $\pi_x \mathbf{1}_{z=x}$, total variation is exactly one minus the correct-decoding probability. Put $u_x = \max_m d(x \mid m)$. Then

$$0 \leq u_x \leq 1, \quad \sum_x u_x \leq K, \quad \mathbb{P}(Z = X) \leq \sum_x \pi_x u_x \leq \sum_{i=1}^{\min(K,N)} \pi_{(i)}.$$

The last inequality follows by moving mass in u from a smaller weight to a larger one until the K largest weights are filled. To attain it, retain distinct labels for those letters and decode them

deterministically; all other letters are mapped to any retained label. Before the cut one may use the unrestricted memory to compute this encoding.

For a hidden selector, condition on its value. It is independent of X , so every conditional correct-decoding probability obeys the same bound; averaging cannot improve it. For a visible selector, variation is the average of the fiber variations, with the same conclusion. The private construction works for both signatures. Finally, the exact joint cut matrix is diagonal with its N positive diagonal entries π_x . Its ordinary rank is N , so its nonnegative rank is at least N and its diagonal factorization attains N . $\square$

The absence of an auxiliary cross-cut feedback channel is essential. An unpriced controller storing X and returning it as a later control would change the experiment, not disprove the cut bound. Likewise a selector correlated with X is not an independent hidden selector. These distinctions are the same ones needed in the collision certificate below.

4. FINITE-DATA TESTING ON THE ORIGINAL BEHAVIOR IMAGE

4.1. The reset interface. Let $B \subset \prod_{j=1}^J \Delta(\Omega_j)$ be compact, and let $Q = (Q_j)$ be fixed. Set $d_j = |\Omega_j|$, $D = \sum_j (d_j - 1)$, $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$, and $f_{j,A}(b) = Q_j(A) - b_j(A)$. Include the empty event. No convexity of B is assumed.

A candidate b is fixed for the whole audit. The auditor obtains n independent samples of each b_j , with independence between rows. In a causal experiment, row j means a specified parameter and deterministic feedback tree; the reset interface supplies that row. It is an additional experimental interface, not information available to the online simulator. A complete sample contains the terminal mark. Private randomness and any per-run hidden selector are reset between runs. The design of the candidate is not resampled.

Write $C_{j,w}$ for the counts, and $\mathcal{N}_n$ for all count arrays with $\sum_w C_{j,w} = n$. Their probability is the polynomial

$$p_C(b) = \prod_j \frac{n!}{\prod_w C_{j,w}!} \prod_w b_j(w)^{C_{j,w}}.$$

A randomized audit is an array $c = (c_{C,i})$ with $c_C \in \Delta(\mathcal{I})$. After training it chooses $i = (j, A)$ according to c_C . Independent candidate and target validation runs give the signed score $\mathbf{1}_A(\text{target}) - \mathbf{1}_A(\text{candidate})$. Its expected value is

$$g_c(b) = \sum_{C,i} p_C(b) c_{C,i} f_i(b), \quad v_n^{\text{aud}}(B, Q) = \max_c \min_{b \in B} g_c(b). \quad (4.1)$$

Thus every audit payoff is polynomial of total degree at most $Jn + 1$ in the behavior coordinates. The probability of a large observed score in a single audit is not asserted to equal its expectation; additional independent validation would be required for a specified confidence level.

Theorem 4.1 (Reset-audit duality and finite-sample approximation). *Let $\delta = \min_{b \in B} \max_i f_i(b)$. Then all the extrema below are attained, and*

$$v_n^{\text{aud}}(B, Q) = \min_{\mu \in \mathcal{P}(B)} \sum_{C \in \mathcal{N}_n} \max_{i \in \mathcal{I}} \int_B p_C(b) f_i(b) d\mu(b), \quad (4.2)$$

$$0 \leq \delta - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}. \quad (4.3)$$

The values are nondecreasing in n . If $a = \dim \text{aff } B$, a minimizing measure in (4.2) can be supported on at most $\binom{a+Jn+1}{a}$ points.

An explicit feasible audit attaining the lower bound $\delta - \sqrt{D/n}$ chooses an event maximizing $Q_j(A) - C_j(A)/n$, with a fixed deterministic tie rule. If the rational target probabilities are given,

this is a rationally specified finite rule. With a free schedule clock, a conservative retained-state bound is

$$H(|\mathcal{I}| + 1)(n + 1)^{\sum_j d_j},$$

where H is the number of states needed to execute and record one causal transcript. This budget is charged to the auditor, not to the simulator.

Proof. The coefficient arrays form a compact convex product of finite simplexes. The integrand is continuous in b and affine in c . The probability measures on compact B form a compact convex set for weak convergence. The minimum over b equals the minimum over such measures for fixed c . Finite-dimensional minimax (or its compact measure version) therefore exchanges the two extrema. Maximization over each independent coefficient simplex selects its largest coordinate integral, giving (4.2). Continuity and compactness give attainment. This step uses measures on the original B , not feasible private realizations of their barycenters.

For any audit, $g_c(b) \leq F(b) = \max_i f_i(b)$. At a minimizer of F this gives $v_n^{\text{aud}} \leq \delta$. The zero event gives $v_n^{\text{aud}} \geq 0$. For the stated explicit rule put $\hat{b}_j = C_j/n$ and $\rho = \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Comparing the empirical maximizer with a true maximizing event gives the pointwise regret bound

$$F(b) - f_{i(C)}(b) \leq 2\rho.$$

For a multinomial empirical measure,

$$\mathbb{E}\|\hat{b}_j - b_j\|_{\text{TV}}^2 \leq \frac{d_j}{4} \sum_w \mathbb{E}(\hat{b}_j(w) - b_j(w))^2 = \frac{d_j(1 - \sum_w b_j(w)^2)}{4n} \leq \frac{d_j - 1}{4n}.$$

Consequently $\mathbb{E}\rho \leq (\sum_j \mathbb{E}\|\hat{b}_j - b_j\|_{\text{TV}}^2)^{1/2} \leq \sqrt{D/n}/2$. Averaging the regret uniformly over B proves (4.3).

From $n + 1$ unordered samples in each row, discard one uniformly at random. The remaining counts have exactly the n -sample law, independently across rows. This realizes every n -sample audit as a randomized $(n + 1)$ -sample audit and proves monotonicity.

Choose affine coordinates on $\text{aff } B$. All the functions $p_C f_i$ lie in the span of monomials of total degree at most $Jn + 1$, a space of dimension at most $s = \binom{a + Jn + 1}{a}$ including the constant. The moment vector of a probability measure is in the convex hull of the evaluation vectors; their constant coordinate is one, so their affine dimension is at most $s - 1$. Eliminating affine dependencies gives a representation by at most s points. It preserves every integral in (4.2), and hence preserves the objective.

Finally, each count is in $\{0, \dots, n\}$. The indicated product register stores all counts, the within-run state, and the selected event for validation. Only the latter event and the within-run buffer need survive once training ends, so the product bound is conservative. Randomizing an event uses fresh randomness. The schedule, reset access and terminal mark readout were declared before the theorem. No private simulator state is read by this implementation. $\square$

The dual measure chooses a candidate once, after which its samples are independent conditional on that candidate. Replacing it by $\bar{b} = \int b d\mu$ would replace $\int p_C(b) f_i(b) d\mu$ by $p_C(\bar{b}) f_i(\bar{b})$, generally a different quantity. It would also replace the joint replicated law by a product of its marginals. Neither operation is part of the theorem. This is the operational meaning of the moment side, rather than a same-budget convexification theorem.

Proposition 4.2 (Both sides sampled). *Suppose the target reference Q^b varies with the candidate, as it does when visible-selector weights are design variables. Supply independent training samples from both b and Q^b , and use their empirical difference. If their row dimensions are D_P, D_Q , the same construction has regret at most $\sqrt{D_P/n} + \sqrt{D_Q/n}$. Minimax and attainment hold on the*

compact image of pairs (b, Q^b) . In particular a variable reference law must not be treated as known fixed input to Theorem 4.1.

Proof. The error in any event discrepancy is at most the sum of the maximal row variations of the two empirical tables. Apply the preceding second moment bound separately and then the empirical-maximizer regret inequality. Count probabilities are products for the two tables. They are continuous polynomials on the compact image of pairs, so the same minimax argument applies without changing the simulator signature. $\square$

4.2. A matching uniform sampling obstruction.

Theorem 4.3 (Binary reset lower bound). *Take one binary row with target $\text{Ber}(1/2)$ and candidate class $B_h = \{\text{Ber}(1/2 - h), \text{Ber}(1/2 + h)\}$, where $0 < h < 1/2$. Then*

$$\delta = h, \quad v_n^{\text{aud}} = h \|\text{Ber}(1/2 + h)^{\otimes n} - \text{Ber}(1/2 - h)^{\otimes n}\|_{\text{TV}}. \quad (4.4)$$

For every $n \geq 1$, choosing $h = 1/(8\sqrt{n})$ gives

$$\delta - v_n^{\text{aud}} \geq \frac{1}{16\sqrt{n}}.$$

Thus the exponent in (4.3) is sharp uniformly over compact candidate classes, already at affine dimension one. This does not assert a degree lower bound for revealed-behavior polynomials, nor that this two-point class equals an unrestricted private-row image.

Proof. Only the two complementary singleton events and the zero event matter. A randomized choice is a statistic d in $[-1, 1]$. The two expected payoffs are $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$. Their minimum is at most half their sum, which is at most $h\|P_+^n - P_-^n\|_{\text{TV}}$. Conversely take the sign of the likelihood difference, assigning zero at a tie. Complementing every sample interchanges P_+^n and P_-^n and changes that sign. The two payoffs are equal, and their average attains the variation bound. The statistic depends only on the count, so it is an admissible audit. The symmetric measure on the two candidates attains the dual value. Every Dirac measure has dual objective h , whereas for finite n the two product laws are equivalent and their variation is strictly less than one. Thus no dual minimizer is a Dirac measure. Their barycenter is the target law, but replacing the dual by that barycenter would incorrectly give value zero instead of the strictly positive value in (4.4).

For $h \leq 1/8$, elementary integration of $(1 - u^2)^{-1}$ gives

$$\text{KL}(\text{Ber}(1/2 + h) \parallel \text{Ber}(1/2 - h)) = 2h \log \frac{1 + 2h}{1 - 2h} \leq 16h^2.$$

The product relative entropy is at most $16nh^2$. Pinsker's inequality bounds its total variation by $\sqrt{8nh^2}$. At the specified h this is less than $1/2$. Substituting in (4.4) proves the claim. The inequality uses the classical two-point testing mechanism; the restriction being measured is the reset budget of the present game. $\square$

4.3. Transport of one fixed audit.

Proposition 4.4 (Reset cost under model perturbation). *Let q range over the same compact set of executable row designs in two models. Suppose uniformly in q, j that*

$$\|P_j^q - \tilde{P}_j^q\|_{\text{TV}} \leq \alpha, \quad \|Q_j - \tilde{Q}_j\|_{\text{TV}} \leq \beta.$$

For the same coefficient rule c , evaluated with each model's own target validation,

$$|g_c(q) - \tilde{g}_c(q)| \leq (2Jn + 1)\alpha + \beta. \quad (4.5)$$

The deficiencies themselves differ by at most $\alpha + \beta$, at the same resource budget. If the source behavior is unchanged, a fixed audit loses at most β , independently of its reset count.

Proof. A product coupling of the Jn training runs fails with probability at most $Jn\alpha$. Taking counts and choosing the event are contractions. A score with fixed target discrepancies lies in $[-1, 1]$, so the change in its expectation from this training-law perturbation is at most $2Jn\alpha$. At each event the discrepancy changes by at most $\alpha + \beta$. This proves (4.5). For deficiency, every row variation changes by at most $\alpha + \beta$; taking the maximum and then the minimum preserves that bound. No new simulator is inserted. $\square$

For a rational polynomial image, a rational rule and a rational threshold, strict positivity of $g_c(q)$ above that threshold is a first-order formula over the original row simplexes. Real quantifier elimination decides it. The same is true of an existential rule followed by this universal condition. This is an exact finite verification statement, not an efficient algorithm. The explicit empirical rule below will be verified by a uniform analytic inequality, rather than by storing its enormous complete table.

5. CONTINUATION COMPLEXITY OF A FINITE-DATA TEST

The reset budget does not by itself bound the auditor's retained memory. We now impose both constraints. At scheduled epoch t a streaming auditor reads one letter x_t from a finite alphabet X_t , updates a register, and emits no information to any auxiliary store. It makes its decision on reading the last letter. Transition rows may depend on the epoch; the clock is free. All other information crossing a cut belongs to the register. Fresh randomization is allowed. An independent persistent selector, when allowed, selects a complete auditor before any letters are read.

For a Boolean decision $\psi : X_1 \times \cdots \times X_n \rightarrow \{-1, 1\}$, let $u \sim_t v$ mean that

$$\psi(us) = \psi(vs) \quad \text{for every } s \in X_{t+1} \times \cdots \times X_n.$$

Write $r_t(\psi)$ for the number of equivalence classes of prefixes of length t , for $0 \leq t < n$. These are the deterministic continuation responses. The terminal decision is emitted on the last transition, so no terminal register is needed. In particular, $r_0 = 1$. The equivalence is a finite-horizon version of the classical residual-state construction [16]. The following quantitative statement will also measure the loss when a statistical auditor has fewer states than this construction.

Theorem 5.1 (Exact continuation width and a quantitative cut obstruction). *A stochastic streaming auditor computes ψ exactly on every word with width profile $(K_t)_{0 \leq t < n}$ if and only if $K_t \geq r_t(\psi)$ at every cut. The profiles are attained simultaneously by a deterministic auditor. Independent persistent randomization does not reduce these widths for exact computation.*

More generally, at a cut with $r = r_t(\psi) \geq 2$, every stochastic auditor of width K satisfies

$$\sum_{w \in X_1 \times \cdots \times X_n} e_\psi(w) \geq \frac{(r - K)_+}{r - 1}, \quad (5.1)$$

where $e_\psi(w)$ is its probability of an incorrect Boolean decision on input w . The inequality also holds after averaging over an independent selector whose conditional auditors each have width at most K .

Proof. From a prefix u let $e_u(m)$ be the distribution of the retained state. From state m and suffix s let $a_m(s)$ be the probability of terminal answer $+1$. Exact computation says

$$\sum_m e_u(m) a_m(s) = \frac{1 + \psi(us)}{2}.$$

The right side is zero or one. Every state of positive e_u -mass must therefore have precisely that zero-one continuation response for all s . Distinct prefix classes have disjoint positive supports. Hence $K_t \geq r_t$. Conversely, retain the prefix class. Appending the same letter to two equivalent prefixes gives equivalent new prefixes; thus every transition is well defined. At the last transition

the class and the last letter determine ψ . This realizes every width r_t simultaneously. For a persistent independent selector, a zero average error on each of the finitely many words implies zero conditional error on all words outside a single null set of selector values. The preceding bound applies there.

For the quantitative claim choose representative prefixes $u_1, \dots, u_r$, one per class, and put $e_i(m) = e_{u_i}(m)$. Define

$$o_{ij} = \sum_{m=1}^K \min\{e_i(m), e_j(m)\}.$$

For nonnegative numbers $a_1, \dots, a_r$, ordering them shows that $\sum_{i < j} \min(a_i, a_j) \geq \sum_i a_i - \max_i a_i$. Apply this at each state. Since every e_i has mass one and each $\max_i e_i(m) \leq 1$, it follows that

$$\sum_{i < j} o_{ij} \geq r - K. \quad (5.2)$$

Choose a suffix s_{ij} distinguishing u_i, u_j . Conditional on a common state and this suffix, the two desired answers are opposite. Their error probabilities, weighted by $e_i(m), e_j(m)$, have sum at least $\min\{e_i(m), e_j(m)\}$. Thus

$$e_\psi(u_i s_{ij}) + e_\psi(u_j s_{ij}) \geq o_{ij}.$$

On summing, each complete word on the left occurs at most $r - 1$ times: its prefix identifies a single representative, which participates in $r - 1$ pairs. Nonnegativity of all other word errors and (5.2) prove (5.1). When $r \leq K$ the claim is just nonnegativity. Finally apply the bound conditional on a selector and integrate. Independence ensures that conditioning has not changed the input experiment. $\square$

The exact part is also the extremal zero-one case of Theorem 3.2. Its content is a simultaneous width profile, rather than separate unrelated factorizations at different cuts. The error bound does not assert that its constant is optimal for every Boolean function. Its use below is to convert a continuation obstruction into a strict loss of the attainable statistical value.

5.1. The width required to attain the binary reset value. Fix $0 < h < 1/2$ and the binary candidate class B_h of Theorem 4.3. A terminal decision $+1$ chooses the singleton event appropriate to the $+$ candidate, and -1 the complementary event. A zero-score event can equivalently be replaced by an independent fair choice of these decisions. The expected payoffs are therefore $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$, where $d \in [-1, 1]$ is the conditional mean decision and $P_\pm = \text{Ber}(1/2 \pm h)^{\otimes n}$. Let $v_{n,K}$ be the supremum of the smaller of these two payoffs over streaming auditors with the declared width profile. The final validation interface is unchanged; the widths here count training and selection only.

Theorem 5.2 (Exact memory threshold for optimal binary auditing). *Let $n = 2m + 1$ be odd. The unconstrained reset value is $v_n = h\|P_+ - P_-\|_{\text{TV}}$. It is attained at the profile*

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n, \quad (5.3)$$

and $v_{n,K} = v_n$ if and only if $K_t \geq r_t$ for all these cuts. For every cut with $r_t \geq 2$ one has the explicit deficit

$$v_n - v_{n,K} \geq 2h^2(1/4 - h^2)^m \frac{(r_t - K_t)_+}{r_t - 1}. \quad (5.4)$$

These conclusions remain valid when an independent persistent selector chooses auditors of the same conditional widths. For $n \geq 3$ the least peak training width attaining v_n is $m + 2$; for $n = 1$ it is one.

Proof. The sign of $P_+(w) - P_-(w)$ is the majority decision $\psi(w)$, with no zero difference since n is odd. Complementing every bit interchanges the two laws and reverses the decision. Consequently majority has equal payoffs and attains $h\|P_+ - P_-\|_{\text{TV}}$, as also proved in Theorem 4.3.

A prefix with k ones requires at least $m + 1 - k$ ones in its remaining $n - t$ places. Thresholds at most zero give the constant positive response; thresholds greater than $n - t$ give the constant negative response; the remaining thresholds give different nonconstant responses. Counting them for $0 \leq k \leq t$ gives (5.3). The exact realization in Theorem 5.1 implements majority at all these widths.

For an arbitrary randomized decision, its smaller payoff is bounded by the equal-prior average. The deficit of this average from the majority value is exactly

$$h \sum_w |P_+(w) - P_-(w)| e_\psi(w). \quad (5.5)$$

Put $a = 1/2 + h$ and $b = 1/2 - h$. A word with $k \geq m + 1$ ones has absolute likelihood difference $(ab)^{n/2} \{(a/b)^{k-n/2} - (a/b)^{-(k-n/2)}\}$, which increases with k . Symmetry treats the other half of the words. The smallest difference is therefore attained at $k = m, m + 1$ and equals

$$a^{m+1}b^m - b^{m+1}a^m = 2h(1/4 - h^2)^m.$$

Substitution of (5.1) into (5.5) gives (5.4). Any deficient cut gives a strictly positive loss; all sufficiently wide profiles implement majority. Maximizing (5.3) proves the peak-width assertion. The same average-payoff argument works conditional on an independent selector and hence after averaging. $\square$

The result identifies the exact memory needed to attain the optimal value, not the memory needed for a prescribed approximate value uniformly in h . The quantitative deficit can be small for large n . In particular it is not a sharp joint asymptotic tradeoff in sample size, state size and error. It complements, rather than replaces, the uniform sampling exponent in Theorem 4.3. No minimax interchange on the generally nonconvex set of fixed-width auditors has been used.

6. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 7.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 2.2, with rational instrument data. Write its objective as $F(q) = \max_{1 \leq i \leq m} f_i(q)$, where f_i are all its marked event discrepancies, including the empty event $f_1 = 0$. For visible selectors the event and feedback rule include their selector fibers, and the target selector law remains in f_i . Thus $-1 \leq f_i \leq 1$ for all three signatures; the polynomials may be different.

Parameterize a row with d outcomes by $d - 1$ variables in $[0, 1]$:

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map $q(x)$ from $[0, 1]^D$ to the feasible row product. Put $h_i(x) = f_i(q(x))$ and $g_i = (1 + h_i)/2$. All h_i are rational polynomials, $0 \leq g_i \leq 1$, and $g_1 = 1/2$. The case $D = 0$ is a single feasible table and is interpreted directly.

Theorem 6.1 (Strict polynomial lower-certificate hierarchy). *For $p \geq 1$ define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0,1]^D} A_p(x), \quad \delta = \min_q F(q).$$

Then the denominators are positive, a_p is nondecreasing, and

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (6.1)$$

For rational L let

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (6.2)$$

A nonnegative tensor Bernstein expansion of $P_{p,L}$ on $[0,1]^D$ certifies $\delta \geq L$. If $L < \delta$, some p and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such expansion certifies $L < \delta$. These are lower certificates at the original state and selector budgets. No convexification of the feasible behavior image occurs.

Proof. Fix x and put $b = \max_i g_i(x) \geq 1/2$ and $r_i = g_i(x)/b$. At least one r_i is one. Write $S_j = \sum_i r_i^j$. Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently $S_{p+1}/S_p \geq m^{-1/(p+1)}$. Since $A_p = 2bS_{p+1}/S_p - 1$ and $\max_i h_i = 2b - 1$,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy-Schwarz gives $S_{p+2}S_p \geq S_{p+1}^2$, so A_p is nondecreasing in p . Its denominator is at least 2^{-p} and it is continuous on the cube. Taking minima proves (6.1). The bound is also at most $2 \log m/(p+1)$, by $1 - e^{-u} \leq u$.

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply $P_{p,L} \geq 0$. Dividing by the positive denominator gives $F(q(x)) \geq A_p(x) \geq L$ for every x , which proves soundness. For completeness choose p so that the error in (6.1) is less than $\delta - L$. Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 6.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most m , so they give a strict positive uniform gap above L . $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk-Laurent [20]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

Lemma 6.2 (An explicit positivity degree). *Let $P(x) = \sum_k c_k x^k$ be a rational polynomial in D variables and $n \geq \max_{k,j} k_j$. In its tensor Bernstein expansion of degree n in each coordinate the coefficient at $\alpha \in \{0, \dots, n\}^D$ is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (6.3)$$

In particular, $\min P > 0$ implies positivity of every coefficient whenever $n > C(P)/\min P$ and the degree restriction holds.

Proof. For a binomial random variable B with parameters (n, x) , $\mathbb{E}\binom{B}{k} = \binom{n}{k}x^k$. Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is $\binom{a}{k} = 0$ for $a < k$. To compare $\binom{a}{k}/\binom{n}{k}$ with $(a/n)^k$, draw k indices uniformly from $\{1, \dots, n\}$. Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most $\binom{k}{2}/n$. Apply this total variation bound to the event that all drawn indices lie in a prescribed a -element subset. Hence the difference of the two factors is at most $\binom{k}{2}/n$. All factors lie in $[0, 1]$. A product telescoping bound followed by the triangle inequality proves (6.3). The final assertion follows by taking the minimum of P . $\square$

Corollary 6.3 (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level L is equivalent to the existence of the following finite rational certificate: an integer p , an integer n , and a strictly positive coefficient array for (6.2) whose identity is checked by (6.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to δ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

Proof. Soundness and completeness are Theorem 6.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all (p, n) and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed (p, n) test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument. $\square$

Remark 6.4 (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions $g_i^p / \sum_j g_j^p$ are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count $(n+1)^D$, the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 7.3; its verification is a different cost.

6.1. A sharp two-time nonconvex example. The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits $(0, 0)$ or $(1, 1)$ with probability $1/2$ each. At width one, a private simulator emits two independent bits with probabilities p, q . It cannot reread its first emitted bit.

Proposition 6.5 (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

The private objective is the maximum of the four polynomials

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1-p)(1-q), \quad p + q - 2pq.$$

Every constant mixture of these tests has minimum at most zero, whereas Corollary 6.3 certifies every rational $L < \sqrt{2} - 1$ at private width one.

Proof. Put $u = pq$ and $v = (1-p)(1-q)$. Direct total variation calculation gives $F = 1 - \min(u, 1/2) - \min(v, 1/2)$ and the displayed maximum. By Cauchy–Schwarz, $\sqrt{u} + \sqrt{v} \leq 1$. If

$u \geq 1/2$, then $v \leq (1 - 1/\sqrt{2})^2$, so the sum of the two capped terms is at most $2 - \sqrt{2}$; the case $v \geq 1/2$ is symmetric. Otherwise set $a = \sqrt{u}$, $b = \sqrt{v}$. On $0 \leq a, b \leq 1/\sqrt{2}$, $a + b \leq 1$, the maximum of $a^2 + b^2$ is at a vertex of this polygon and equals $2 - \sqrt{2}$. Equality is attained at $p = q = 1/\sqrt{2}$. A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation. $\square$

7. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

7.1. The testing game. Let B be a nonempty compact subset of a finite-dimensional affine space, and let $f_1, \dots, f_m$ be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 2, b is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts; f_i ranges over their signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set B . No convex hull is taken.

An *adaptive test* is a continuous map $\kappa : B \rightarrow \Delta_m$, chosen before a candidate b . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function κ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

Theorem 7.1 (Revealed-behavior adaptive representation). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{7.1}$$

The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.

Proof. Fix $\eta > 0$. The sets $U_i = \{b : f_i(b) > F(b) - \eta\}$ form an open cover of the compact metric space B . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has $F(b) - \eta < G(b, \kappa) \leq F(b)$ for all b . Consequently the first supremum equals δ . The same uniform inequality shows, for every probability μ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is δ , attained by a point mass at a minimizer of F . This also proves the displayed minimax exchange without an assumption that B is convex. Equivalently one can apply the usual compact-convex minimax theorem to $\mathcal{P}(B)$ and the convex class of continuous tests, with the pairing $\int G d\mu$. Finally choose the smallest index attaining $F(b)$. Its level sets are Borel, and its indicator vector is a Borel test with score exactly F . $\square$

The probability μ here describes a lottery over *revealed candidate laws*. The score is $\int G(b, \kappa) d\mu(b)$, not $G(\int b d\mu, \kappa)$. Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

7.2. A size bound in behavior dimension. Factor out directions annihilated by every $f_i - f_i(b_0)$, so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by a and let $R = \text{diam}_*(B)$. All conclusions below are unaffected by this observable quotient. The case $a = 0$ is a single constant comparison and requires one test. Suppose henceforth that $a \geq 1$, so $R > 0$.

Lemma 7.2 (A bounded behavior chart). *There exist $b_0 \in B$ and vectors $v_j \in B - b_0$, $1 \leq j \leq a$, forming a basis, such that $\|v_j\|_* \leq R$ and every $b \in B$ has coordinates $b = b_0 + \sum_j t_j v_j$ with $|t_j| \leq 2$. If B is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

Proof. Fix $b_0 \in B$. In any fixed coordinates on the difference space, let $D_0 > 0$ be the maximum absolute determinant of a vectors in $B - b_0$. Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than $D_0/2$. Replacing its j th vector by $b - b_0$ multiplies the determinant by t_j ; maximality gives $|t_j| < 2$. The norm bound is the definition of R . For a rational polynomial image, take b_0 at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart. $\square$

Set $b(x) = b_0 + \sum_j (4x_j - 2)v_j$ and $C = \{x \in [0, 1]^a : b(x) \in B\}$. The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write $\ell_i(x) = f_i(b(x))$. Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (7.2)$$

For $n \geq 0$ let $\beta_{\alpha, n}$ be the tensor Bernstein basis of coordinatewise degree n . A feasible coefficient array has $c_{\alpha i} \geq 0$ and $\sum_i c_{\alpha i} = 1$ for every $\alpha \in \{0, \dots, n\}^a$. It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha, n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha, n}(x) \ell_i(x). \quad (7.3)$$

Theorem 7.3 (Intrinsic adaptive degree and support bounds). *The maxima in (7.3) are attained and $d_n \leq d_{n+1} \leq \delta$. For $n \geq 1$,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (7.4)$$

There is an array with at most $(n+1)^a$ nonzero coefficients, all equal to one, whose minimum score is at least $\delta - 4Ra/\sqrt{n}$. Its test has coordinatewise degree n , and its score degree at most $n+1$. Furthermore

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (7.5)$$

and a minimizing measure can be supported on at most $(n+2)^a$ points. In particular a strict level $L < \delta$ admits an adaptive test of positive uniform margin with $n > (8Ra/(\delta - L))^2$ and at most $(n+1)^a$ occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.

Proof. The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact C is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the $\ell_i(x)$, proving the upper bound by δ .

At each grid point α/n choose an index i_{α} maximizing ℓ_i there. The choice is made even at infeasible grid points; only the final score is minimized over C . Give that index coefficient one and the others zero. For fixed x let A_j be independent $\text{Bin}(n, x_j)$ variables. If i_* maximizes at x ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses $\text{Var}(A_j/n) \leq 1/(4n)$. This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over C by a minimum over $\mathcal{P}(C)$. The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [18] exchanges the extrema. Optimizing each coefficient simplex gives (7.5). The integrands are polynomials of coordinatewise degree at most $n+1$. Preserve all monomial moments of that degree by Carathéodory's theorem. There are $(n+2)^a$ monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most $(n+2)^a - 1$. A measure with at most $(n+2)^a$ atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than $(\delta - L)/2$. $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (7.5) cannot be replaced by its behavior barycenter. For $n=0$ it can: d_0 is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 6.5, the target belongs to the convex hull of private behaviors, so $d_0 = 0$, whereas $\lim_n d_n = \sqrt{2} - 1$. Thus the hierarchy recovers a gap which every constant test misses.

Remark 7.4 (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set C , the bit lengths of a rational chart, or the size of a description of C . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use $R \leq 1$ with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound $R \leq 2$ is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

7.3. A three-weight certificate invisible to constant tests.

Proposition 7.5 (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities p, q , for the target which assigns probability $1/2$ to each of 00, 11. Put $u = pq$, $v = (1-p)(1-q)$ and $J(t) = 3t^2 - 2t^3$. The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

are nonnegative and sum to one. Against the event discrepancies $1/2 - u, 1/2 - v, 1 - u - v$, respectively, their score G satisfies

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (7.6)$$

The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.

Proof. We have $u + v \leq 1$. Since J is increasing on $[0, 1]$ and $J(1-t) = 1 - J(t)$, the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set $a = (p + q - 1)^2$ and $b = (p - q)^2$. Then $a, b \geq 0$ and $a + b \leq 1$. Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let $c = 1 - a - b$. Another polynomial identity, valid when $a + b + c = 1$, is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus $G(a, b) \geq G(a, 0)$. The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients $z_k/349440$, where

$$\begin{aligned} (z_0, \dots, z_{16}) &= (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ &\quad 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944). \end{aligned}$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is $351/349440 = 9/8960$, proving (7.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero. $\square$

For comparison, the affine weights $v, u, 1 - u - v$ already give the smaller bound $3/8$, with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

The infinite-dimensional representation above holds for continuous f_i on any compact metric space: its proof uses a partition of unity, not causal factorization. The measure minimum is attained at a Dirac mass at a minimizer of F . The finite-level polynomial theorem has a different

content. It is an existence-size bound for revealed-law verification; the reset-audit theorem of Section 4 imposes an explicit finite-data restriction.

7.4. Strict certificates and effective physical input. Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at $L = \delta$.

Proposition 7.6 (Uniform effective presentation input). *Suppose a single algorithm, on input j , returns rational finite instruments E_j, F_j and nonnegative rational error enclosures a_j, c_j for two-sided stateless marked feedback presentations of E, F . Suppose $a_j + c_j \rightarrow 0$. All alphabets, widths, selector laws and finite subroutines must be computable uniformly from j and the requested signature. Then, for every positive rational η , an algorithm returns a rational interval of width below η containing $\delta_b(E, F)$ at the same signature.*

Proof. Enumerate j until $a_j + c_j < \eta/8$. This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below $\eta/2$. Enlarge it by $a_j + c_j$ at both ends using stateless comparison. The enlarged width is below $3\eta/4$. Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition. $\square$

8. AN EXACT ALGORITHM FOR THE OBSERVABLE CHART

The chart in Lemma 7.2 can be found without taking its dimension as an oracle. We state the input and output explicitly. Rational polynomials are encoded by finite lists of monomial exponent vectors and rational coefficients in reduced integer form. A product of simplexes is encoded by the number of entries in each row. Eliminating the last entry of every nontrivial row leaves a full-dimensional rational polytope $\Omega \subset \mathbb{R}^r$. Zero-dimensional factors are omitted.

Proposition 8.1 (Effective observable quotient and pullback). *The input consists of the preceding simplex description, a rational polynomial map $b : \Omega \rightarrow \mathbb{R}^N$, and finitely many rational affine functionals f_i on $\mathbb{R}^N$. There is a terminating algorithm which returns its observable affine dimension a , a rational basis and coordinate map for the observable quotient, and, when $a > 0$, rational parameter points $q_0, q_1, \dots, q_a$ with the following properties. With*

$$z(q) = (f_i(b(q)) - f_i(b(q_0)))_i, \quad v_j = z(q_j),$$

the vectors v_j are a basis of the observable difference space, and all coordinates $t(q)$ in that basis satisfy $|t_j(q)| < 2$. The output also contains the rational polynomial cube map $x(q) = (t(q) + 2)/4$, a rational upper bound for the observable diameter, and the exact pullback of any subsequently supplied rational polynomial in x . The algorithm outputs the universal determinant formula it has decided. It makes no polynomial-time or coefficient-height claim.

Proof. Choose a rational relative interior point q_0 of the input simplexes. Expand the vector polynomial $z(q)$ in the free row coordinates. Let V be the rational span of all its coefficient vectors, including its constant coefficient. This equals $\text{span } z(\Omega)$: a linear functional vanishes on $z(\Omega)$ exactly when its polynomial vanishes on the nonempty interior of Ω , hence identically, hence on

every coefficient. Gaussian elimination therefore gives $a = \dim V$ and a rational basis matrix U . Choosing independent rows gives a rational left inverse L with $LU = I_a$. Thus $y(q) = Lz(q)$ provides exact quotient coordinates. If $a = 0$, all discrepancies are constant and no chart is required.

For $a > 0$ let

$$D_0 = \max_{u_1, \dots, u_a \in \Omega} |\det(y(u_1), \dots, y(u_a))|.$$

Compactness gives a maximum, and spanning gives $D_0 > 0$. Enumerate rational tuples $(q_1, \dots, q_a)$ in Ω^a . Membership and their determinant d are decided by rational arithmetic. For $d \neq 0$, decide the first-order real formula

$$\forall u_1, \dots, u_a \in \Omega : \quad \det(y(u_1), \dots, y(u_a))^2 < 4d^2. \quad (8.1)$$

Quantifier elimination terminates for each formula. Density of rational points and continuity of determinant imply that some enumerated tuple has $|d| > D_0/2$. For such a tuple the strict formula is true, so the search terminates. No computation of an exact algebraic maximizer is needed.

Invert the rational matrix with columns $y(q_j)$ to obtain $t(q)$. Cramer's rule and (8.1) give $|t_j(q)| < 2$ for every q . In the observable norm $\|z\|_* = \max_i |z_i|$, each v_j has norm at most the diameter of $z(\Omega)$. An upper bound for that diameter can be produced, for example, by bounding each free coordinate in $[0, 1]$ and summing the absolute polynomial coefficients for $f_i(b(q)) - f_i(b(q'))$. This bound is rational and computable. Rational matrix multiplication and substitution give $x(q)$ and every requested pullback. They also give its full sparse coefficient encoding. All searches have now been specified by finite encodings and terminating operations. $\square$

There is a small but important distinction between the quotient and a projection onto a smaller set of simulator rows. The former identifies exactly the directions invisible to the chosen event family; the original row constraints remain present in every universal positivity test. The chart need not identify equivalent tasks after a change of marks or of information signature. Those are changes to the comparison problem.

9. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

9.1. The exact marked two-report experiment. Let $B, W \in \{0, 1\}$ have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (9.1)$$

The blind source E_* has two constant reports and actual mark W . The target F_* reports $Y_1 = B$, receives $A \in \{0, 1\}$ after that report, and reports $Y_2 = AB$. The feedback tester may choose A as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate A . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of (B, W) . Put

$$d_* = \sqrt{5/2} - 9/8.$$

Theorem 9.1 (An exact marked causal separation). *At the width profile $(1, 1)$,*

$$\delta^p(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (9.2)$$

The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter $\sqrt{5/8}$, followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.

Proof. First choose the constant gate $A = 1$. The two private emissions are independent Bernoulli variables with parameters p, q and are independent of W . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is $H(u) + H(v)$, where

$$H(t) = \frac{1}{2} \{ \min(t, a) + \min(t, b) \}.$$

Indeed the target is supported on equal report pairs, and its two W -masses at each equal pair are $3/16$ and $5/16$. Total variation is one minus this overlap.

Cauchy-Schwarz gives $\sqrt{u} + \sqrt{v} \leq 1$. By symmetry suppose $u \geq v$. If $u \leq 1/4$ the overlap is at most $1/2$. If $u \geq 1/4$, then $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$, so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of $[1/4, a]$ and $[a, b]$ and nonincreasing on $[b, 1]$. Its maximum is therefore attained at $1/4$, a or b . The values at a and b are $1 + 2a - 2\sqrt{a}$ and $3/2 + b - 2\sqrt{b}$ respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using $\sqrt{a} > 3/5$, $\sqrt{b} > 3/4$. It also exceeds $1/2$. Taking $p = q = \sqrt{b}$ attains this maximum, and hence the minimum error is $1 - (3/2 + b - 2\sqrt{b}) = d_*$.

For any hidden selector, the all-on report pair is still independent of W . If its probabilities of 11 and 00 are u, v , then $u + v \leq 1$. Concavity and monotonicity of H give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit J , emitting J twice, attains this overlap. This requires precisely two deterministic selector tables and gives error $1/8$. In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair (X_1, X_2) , emit X_1 , and after receiving A emit AX_2 . For a deterministic feedback $A = g(X_1)$ the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the private case, and the bit J is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of (B, W) . The all-on gate in every fiber has error at least d_* in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains d_* . $\square$

9.2. A full-dimensional collision preparation. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$ with contact distance d . The center of mass $Q = (q_1 + q_2)/2$ ranges over any fixed rational position box. Write $r = q_1 - q_2$, $g = v_1 - v_2$, and $V = (v_1 + v_2)/2$. Conditional on (B, W) in (9.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], & (9.3) \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

The boxes have positive volume in the full phase space. The actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

Lemma 9.2 (Uniform transverse collision signal). *For every preparation in (9.3) and every $d \in [9/10, 11/10]$, exactly one collision occurs before time 3. Its time t_c lies in $(4/5, 3/2)$, and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (9.4)$$

For $0 < d \leq 2/5$ no collision occurs before time 3, and the detector is zero throughout that interval.

Proof. Up to a collision $r(t) = r + tg$. For $t \leq 4/5$, its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At time $3/2$ the absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. Thus a first contact occurs in the stated interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$ as well. The first contact therefore has $n_x < 0$, where $n = r(t_c)/d$. The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing, $r(t_c) \cdot g' > 0$, so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (9.4). For $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. Thus there is no collision. The incoming $|v_{1,y}| \leq 3/200$ then makes h zero. $\square$

At time 2 the physical target $F_{d,\sigma}$ reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma \xi_1 > 0\}}.$$

It then receives a gate $A \in \{0, 1\}$, and at time 3 reports

$$Y_2 = A \mathbf{1}_{\{h(v_{1,y}(3)) + \sigma \xi_2 > 0\}},$$

where the ξ_i are independent standard normal variables, independent of the preparation. The source E_d has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition

experiment: taking $A = 0$ suppresses later information and taking $A = 1$ exposes another noisy observation of a collision-generated quantity.

Theorem 9.3 (Collision-sensitive physical resource separation). *Let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/12$. At retained width $(1, 1)$ the physical deficiencies obey, uniformly in d ,*

$$\begin{aligned} |\delta^p(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^v(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (9.5)$$

In particular private and visible error exceed $9/20$, while hidden two-selector error is less than $13/100$. More generally the errors in these three comparisons are at most $2\Phi(-1/(4\sigma))$, which tends to zero as $\sigma \downarrow 0$. For $0 < d \leq 2/5$ all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.

Proof. Couple the ideal and physical target by the same (B, W) and the same feedback random table. Lemma 9.2 bounds each potential bit error by $\Phi(-1/(4\sigma))$. With probability at least $1 - 2\Phi(-1/(4\sigma))$ both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly E_* . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$ at $x = 3$ gives $\Phi(-3) < 1/600$: use $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$. Hence the total error is below $1/300$. The inequalities $d_* - 1/300 > 9/20$ and $1/8 + 1/300 < 13/100$ follow by squaring positive rational bounds.

When $d \leq 2/5$, both potential reports are independent fair noise bits and are independent of W . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under $(q, d, t) \mapsto (sq, sd, st)$ with v unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit. $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

9.3. Uniformly effective finite presentations. The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

Proposition 9.4 (Collision-chart rational presentations). *For computable $d \in [9/10, 11/10]$ and $\sigma > 0$, one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for $(E_d, F_{d,\sigma})$. On a compact computable parameter interval with $\sigma \geq \sigma_0 > 0$, the errors and the construction are uniform. Thus Proposition 7.6 applies at every fixed finite state and selector signature, not only at $(1, 1)$.*

Proof. On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals $-d(g \cdot n) > (9/10)(1443/1000)$. All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound C_{geo} for $v'_{1,y}$ in the packet variables and d . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability $p = \Phi(h(v'_{1,y})/\sigma)$. Their four probabilities are p^2 , $p(1-p)$, $p(1-p)$ and $(1-p)^2$. The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most $2C_{\text{geo}}/\sigma_0$ for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh h therefore have explicit error at most a computable constant times h , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (9.1). For each W , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$ exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence. $\square$

10. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let Z take values in a standard Borel space and let Y take values in a standard Borel canonical path space on $[0, T]$, with a right-continuous observation filtration. Consider equivalent joint probability measures P, Q on (Z, Y) , with the same Z -marginal ν . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable $f(Z)$ write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include T and can be evaluated on a countable dense time set.

Theorem 10.1 (Marked variation controls the whole posterior process). *Let $\varepsilon = \|P - Q\|_{\text{TV}}$ and $|f| \leq 1$. When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (10.1)$$

There is a coupling of (Z, Y^P) and (Z, Y^Q) which retains exactly the same latent variable and satisfies

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (10.2)$$

The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.

Proof. Let $D = dP/dQ$, $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$, and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since $\mathbb{E}_Q|D - 1| = 2\varepsilon$, both martingales A, B have terminal absolute expectation at most 2ε . Doob's weak L^1 inequality gives, for $0 < u \leq 2$,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating $2u$ times this bound on $(0, 2)$ gives $16\varepsilon + 64\varepsilon = 80\varepsilon$, proving (10.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on Z , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the event of identical full output paths use (10.1) with the Q -marginal; on its complement the squared difference is at most four. This proves (10.2). The maximal coupling may depend on the full output path; it has no asserted online implementation. $\square$

Corollary 10.2 (Changing-filtration convergence from marked comparison). *Let P_n and P be equivalent joint laws with the same latent marginal ν , and $\|P_n - P\|_{\text{TV}} \rightarrow 0$. They admit a common probability-space realization $(Z, Y, Y^1, Y^2, \dots)$ such that, for every $f \in L^2(\nu)$,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for $f \in L^1(\nu)$. It also permits $f_n \rightarrow f$ in $L^2(\nu)$, respectively in $L^1(\nu)$, in the first conditional expectation.

Proof. Glue the conditional maximal couplings of (Z, Y^n) with the fixed reference (Z, Y) by taking their conditional kernels independently on a countable product. The coupling does not depend on f . For the truncation $f^{(K)} = (-K) \vee (f \wedge K)$, Theorem 10.1 bounds the middle difference in the L^2 norm of the supremum by $K\sqrt{84\varepsilon_n}$. Doob's L^2 inequality bounds each truncation tail by $2\|f - f^{(K)}\|_{L^2(\nu)}$. The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\varepsilon_n}.$$

First let $n \rightarrow \infty$, then $K \rightarrow \infty$. For L^1 use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference $f_n - f$ is handled by the same maximal inequalities under P_n , whose latent marginal remains ν . $\square$

10.1. A microscopic sufficient condition. Let $(\Phi_t)_{t \in \mathbb{R}}$ be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability μ . Fix a prepared law ν with $d\nu/d\mu \leq C$, and bounded real observables h, h_n . These may be collision-sensitive tagged observables. Put $\eta_n = \|h_n - h\|_{L^2(\mu)}$. With a common latent initial state $Z \sim \nu$, consider the continuous observation laws

$$Y_t = \int_0^t h(\Phi_s Z) ds + \sigma B_t,$$

$$Y_t^n = \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0,$$

where the Brownian noise is independent of Z in each model. The notation does not prescribe a synchronous coupling.

Theorem 10.3 (Fixed-flow observation-approximation limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (10.3)$$

For every $|f(Z)| \leq 1$ they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n^2. \quad (10.4)$$

In particular $\eta_n \rightarrow 0$ supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when h_n are orbit-mollified graph-core approximants to any bounded $h \in L^2(\mu)$.

Proof. Conditional on Z , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses $\nu \leq C\mu$ and invariance of μ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (10.3). Applying Theorem 10.1 proves (10.4).

For the last assertion choose a nonnegative smooth approximate identity ρ_n on the time line and set $h_n = \int \rho_n(s) h \circ \Phi_s ds$. Strong continuity of the Koopman group in $L^2(\mu)$ gives $\eta_n \rightarrow 0$ and $\|h_n\|_\infty \leq \|h\|_\infty$. Repeated integration by parts puts h_n in every power domain of the generator, with derivatives obtained from $\rho_n^{(k)}$. This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate. $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (10.3). That is the resource-comparison consequence. The latent-preserving coupling in (10.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

11. CHANGING TRAJECTORIES, LIKELIHOODS, AND CONDITIONAL CONSUMERS

11.1. A common preparation, not a common flow. Let $(\mathcal{Z}, \nu)$ be a standard Borel probability space, and let $H, H_n : [0, T] \times \mathcal{Z} \rightarrow \mathbb{R}$ be jointly measurable, with $|H|, |H_n| \leq M$. They may be observations of different flows on different phase spaces, pulled back by specified preparation maps from $\mathcal{Z}$. No common microscopic generator or invariant measure is assumed. Set

$$\eta_n^2 = \int_{\mathcal{Z}} \int_0^T |H_n(t, z) - H(t, z)|^2 dt d\nu(z).$$

On $\mathcal{Z} \times C([0, T])$ let R be the law of independent $Z \sim \nu$ and $Y = \sigma B$, where $\sigma > 0$. Define

$$L_t^n = \exp \left\{ \frac{1}{\sigma} \int_0^t H_n(s, Z) dB_s - \frac{1}{2\sigma^2} \int_0^t H_n(s, Z)^2 ds \right\}, \quad (11.1)$$

$$dP_n = L_T^n dR. \quad (11.2)$$

The reference model P, L uses H . Conditional on Z , these are Gaussian shifts, so under P_n the observed path has law $Y_t = \int_0^t H_n(s, Z) ds + \sigma B_t^n$. Boundedness gives true strictly positive density martingales. Write $\mathcal{F}_t$ for the usual canonical observation filtration and $\mathcal{H}_t = \sigma(Z, B_s : s \leq t)$ for the larger reference filtration. Equivalent laws have the same completed observation stopping times.

Theorem 11.1 (Likelihood and posterior transport for changing paths). *The measures P_n, P, R are equivalent and have common Z -marginal ν . With $A = M^2 T / \sigma^2$ and $\varepsilon_n = \min\{1, \eta_n / (2\sigma)\}$, one has*

$$\|P_n - P\|_{\text{TV}} \leq \varepsilon_n, \quad (11.3)$$

$$\mathbb{E}_R \sup_{t \leq T} |L_t^n - L_t|^2 \leq \frac{8}{\sigma^2} e^{3A} \eta_n^2. \quad (11.4)$$

For $|f(Z)| \leq K$ put

$$N_t^{n,f} = \mathbb{E}_R[L_T^n f(Z) | \mathcal{F}_t], \quad \ell_t^n = N_t^{n,1}.$$

Then

$$\mathbb{E}_R \sup_t |N_t^{n,f} - N_t^f|^2 \leq \frac{8K^2}{\sigma^2} e^{3A} \eta_n^2. \quad (11.5)$$

In particular this applies to the projected likelihood ℓ^n . For every $|f| \leq 1$, the posterior functions satisfy the same-path bound $80\varepsilon_n$ of Theorem 10.1, and there is one latent-preserving coupling, independent of f , on which

$$\mathbb{E} \sup_t |\mathbb{E}_{P_n}[f(Z) | \mathcal{F}_t^{Y^n}] - \mathbb{E}_P[f(Z) | \mathcal{F}_t^Y]|^2 \leq 84\varepsilon_n. \quad (11.6)$$

If $\eta_n \rightarrow 0$ and deterministic microscopic paths $X_n(Z) \rightarrow X(Z)$ in a specified Polish path metric in ν -probability, then on this coupling the microscopic paths, the full and projected likelihood processes evaluated on their own observation paths, and each bounded posterior process converge jointly in probability in that metric and uniformly in time for the last three coordinates.

Proof. Conditional Gaussian change of measure gives

$$\text{KL}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |H_n - H|^2 dt = \frac{\eta_n^2}{2\sigma^2}.$$

The same calculation proves conditional equivalence, and the conditional density has mean one, so the preparation marginal is unchanged. Integrating and applying Pinsker proves (11.3).

For the quantitative density estimate, condition first on $Z = z$ and write $D_t = L_t^n - L_t$. The stochastic differential equation is

$$dD_t = \sigma^{-1} \{D_t H_n(t, z) + L_t(H_n - H)(t, z)\} dB_t.$$

The conditional second moment of L_t is at most e^A . Itô's isometry and $(u + v)^2 \leq 2u^2 + 2v^2$ therefore imply

$$\mathbb{E}_R[D_t^2 \mid z] \leq \frac{2M^2}{\sigma^2} \int_0^t \mathbb{E}_R[D_s^2 \mid z] ds + \frac{2e^A}{\sigma^2} \int_0^t |H_n - H|^2(s, z) ds.$$

Gronwall, followed by integration in z , gives $\mathbb{E}_R D_T^2 \leq (2/\sigma^2) e^{3A} \eta_n^2$. The difference is a square-integrable $\mathcal{H}_t$ -martingale. Doob's L^2 inequality proves (11.4). Applying the same maximal inequality to the $\mathcal{F}_t$ -martingale $\mathbb{E}_R[(L_T^n - L_T)f \mid \mathcal{F}_t]$ proves (11.5). These are direct process estimates, not a claim that finite-dimensional convergence alone implies tightness.

The posterior assertions follow from Theorem 10.1 and its countable conditional-coupling construction. For the last claim, on the event $Y^n = Y$ the two likelihood functions are evaluated on the same (Z, Y) . Their uniform difference tends to zero in R -probability by the estimates just proved, and therefore in P -probability by absolute continuity of the fixed P with respect to R . The complementary event has probability at most ε_n . The same argument applies to ℓ^n , and (11.6) handles the posterior. The microscopic convergence holds on any coupling retaining the same Z . A finite union bound proves joint convergence. Stochastic integral versions are chosen measurably on the canonical product space; equivalence makes these common versions valid under every law in the countable sequence. $\square$

For $f \in L^2(\nu)$ the posterior conclusion follows by truncation exactly as in Corollary 10.2; it also permits $f_n \rightarrow f$ in $L^2(\nu)$. Boundedness of H_n is being used for the uniform likelihood second moments. It is not replaced by an unverified uniform-integrability assertion in a singular or zero-noise limit.

11.2. The projected stochastic exponential. Let $\hat{H}_t^n = \mathbb{E}_{P_n}[H_n(t, Z) \mid \mathcal{F}_t]$, using a predictable version for Lebesgue-almost every t . Put $\theta_t^n = \hat{H}_t^n / \sigma$.

Proposition 11.2 (Identification before passage to the limit). *The projected likelihood and the innovation satisfy*

$$\ell_t^n = \exp \left\{ \int_0^t \theta_s^n dB_s - \frac{1}{2} \int_0^t (\theta_s^n)^2 ds \right\}, \quad (11.7)$$

$$I_t^n = B_t - \int_0^t \theta_s^n ds. \quad (11.8)$$

Here I^n is a P_n -Brownian motion in $\mathcal{F}_t$. In particular the limit of the projected likelihoods in (11.5) is the projected stochastic exponential of the limiting model. Change of measure commutes with this limit on bounded measurable path tests.

Proof. Project $L_t^n = 1 + \int_0^t L_s^n H_n(s, Z) dB_s / \sigma$ onto the Brownian observation filtration. The integrands are square integrable; testing against bounded elementary predictable stochastic integrals and using Itô's isometry gives

$$d\ell_t^n = \sigma^{-1} \mathbb{E}_R[L_t^n H_n(t, Z) \mid \mathcal{F}_t] dB_t = \ell_t^n \theta_t^n dB_t.$$

The last equality is Bayes' formula. Since $|\theta_t^n| \leq M/\sigma$, the solution is the positive stochastic exponential (11.7); Girsanov gives the stated innovation. The same argument with H identifies the limiting exponential. Finally $|\mathbb{E}_{P_n} F - \mathbb{E}_P F| \leq 2\|F\|_\infty \varepsilon_n$ for bounded signed F . There is no interchange based only on a formal limit of logarithms. $\square$

11.3. A bounded entropic backward-equation consumer. Fix $\gamma > 0$ and a latent payoff f with $|f| \leq 1$. Define

$$U_t^n = \frac{1}{\gamma} \log \mathbb{E}_{P_n}[e^{\gamma f(Z)} \mid \mathcal{F}_t] = \frac{1}{\gamma} \log \frac{N_t^{n, e^{\gamma f}}}{\ell_t^n}. \quad (11.9)$$

The terminal condition here is the observable conditional certainty equivalent U_T^n , not the generally unobserved $f(Z)$ itself. This specification avoids imposing a latent terminal value on the smaller observation filtration.

Theorem 11.3 (Entropic values and localized backward integrands). *There is a unique bounded entropic value process with terminal value U_T^n ; for a predictable integrand Z^n it satisfies*

$$U_t^n = U_T^n + \int_t^T \frac{\gamma}{2} |Z_s^n|^2 ds - \int_t^T Z_s^n dI_s^n. \quad (11.10)$$

On the coupling of Theorem 11.1,

$$\mathbb{E} \sup_t |U_t^n(Y^n) - U_t(Y)|^2 \leq 84 \left(\frac{e^{2\gamma} - 1}{\gamma} \right)^2 \varepsilon_n. \quad (11.11)$$

For each $r > e^\gamma$ let τ_r^n be the first exit of any of $\ell^n, \ell, N^{n,e^{\gamma f}}, N^{e^{\gamma f}}$ from $(1/r, r)$, capped at T . If $\eta_n \rightarrow 0$, then

$$\mathbb{E}_R \int_0^{\tau_r^n} |Z_s^n - Z_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^{\tau_r^n} |\theta_s^n - \theta_s|^2 ds \rightarrow 0. \quad (11.12)$$

Furthermore $\lim_{r \rightarrow \infty} \sup_n R(\tau_r^n < T) = 0$. Thus the localizations exhaust the observation experiment in probability.

Proof. For a positive bounded latent g , write $N_t^{n,g} = \mathbb{E}_R[L_T^n g \mid \mathcal{F}_t]$ and $dN_t^{n,g} = a_t^{n,g} dB_t$. Projection of the full integral gives

$$a_t^{n,g} = \sigma^{-1} \mathbb{E}_R[L_t^n g H_n(t, Z) \mid \mathcal{F}_t], \quad |a_t^{n,g}| \leq (M/\sigma) N_t^{n,g}.$$

The same representation with $g = 1$ gives $d\ell_t^n = b_t^n dB_t$ and $b_t^n / \ell_t^n = \theta_t^n$. Put $N^n = N^{n,e^{\gamma f}}$ and $a^n = a^{n,e^{\gamma f}}$. Itô's formula in (11.9) gives the martingale coefficient

$$Z_t^n = \gamma^{-1} (a_t^n / N_t^n - b_t^n / \ell_t^n).$$

Replacing dB_t by $dI_t^n + \theta_t^n dt$ reduces its drift to $-\gamma |Z_t^n|^2 / 2$, which proves (11.10). The ratios are bounded by M/σ , so the integrals are well defined. Conversely, exponentiating any bounded solution of (11.10) produces a bounded P_n -martingale with terminal value $e^{\gamma U_T^n}$. Its conditional-expectation representation is unique, and martingale representation then identifies its integrand. This proves uniqueness in the bounded class.

Apply Theorem 10.1 to $g_0 = (e^{\gamma f} - e^{-\gamma}) / (e^\gamma - e^{-\gamma}) \in [0, 1]$. The map taking its conditional expectation to $\gamma^{-1} \log \mathbb{E}[e^{\gamma f} \mid \mathcal{F}_t]$ has Lipschitz constant at most $(e^{2\gamma} - 1)/\gamma$. This proves (11.11).

The initial values of N^n and N agree, since Z has the same marginal. Itô's isometry and the terminal version of (11.5) imply

$$\mathbb{E}_R \int_0^T |a_s^n - a_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^T |b_s^n - b_s|^2 ds \rightarrow 0.$$

Before τ_r^n , all denominators lie in $[1/r, r]$, and the limiting numerators a, b are bounded in absolute value by $(M/\sigma)r$. For example,

$$\left| \frac{a^n}{N^n} - \frac{a}{N} \right| \leq r |a^n - a| + (M/\sigma) r^3 |N^n - N|.$$

The supremum estimates (11.5), followed by integration in time, prove (11.12). The same inequality applies to b^n / ℓ^n .

Finally $e^{-\gamma} \ell^n \leq N^n \leq e^\gamma \ell^n$. Each ℓ^n is a stochastic exponential with uniformly bounded integrand θ^n . Its positive and negative moments of the supremum are bounded uniformly in n : write any power as a stochastic exponential times its bounded compensator and apply Doob's inequality to a higher positive power. The same calculation applies to its inverse. Markov's

inequality, the preceding comparison of N^n with ℓ^n , and a union bound prove the exhaustion assertion. This is a localized integrand theorem; it does not assert an unspecified global topology for backward equations with unbounded terminal data or different nonlinear drivers. $\square$

Corollary 11.4 (A changing-filtration stopping consumer). *Let $\psi(t, x) \in [0, 1]$ be measurable and L_ψ -Lipschitz in x , uniformly in t . For a bounded latent f with $|f| \leq 1$, put $M_t^n = \mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t]$. Then*

$$\left| \sup_{\tau \leq T} \mathbb{E}_{P_n} \psi(\tau, M_\tau^n) - \sup_{\tau \leq T} \mathbb{E}_P \psi(\tau, M_\tau) \right| \leq \varepsilon_n + L_\psi \sqrt{80\varepsilon_n}.$$

The same statement for the entropic value U has the additional Lipschitz factor $(e^{2\gamma} - 1)/\gamma$ on the square-root term. The suprema range over the common canonical stopping times; attainment is not needed for the estimate.

Proof. For a fixed stopping time, first change the measure while keeping its $[0, 1]$ reward function, at cost at most ε_n . Then compare the two reward functions on the same canonical path under P , using Theorem 10.1, Cauchy–Schwarz and the Lipschitz bound. The estimate is uniform in the stopping time, so it survives both suprema. Equivalence of the observation laws is what allows the same stopping-time class. The entropic assertion follows from the Lipschitz calculation in the preceding theorem. $\square$

11.4. Resource transport is not the posterior coupling. Apply any common finite causal acquisition map to Y and retain a fixed finite mark $w(Z)$. Data processing makes its marked row variations at most ε_n , uniformly over a declared finite feedback family. If the source and target have respective bounds α_n, β_n , Theorem 2.2 gives same-budget deficiency perturbation at most $\alpha_n + \beta_n$, and Proposition 4.4 transports a fixed reset audit. A subsequent consumer of width S composed with a simulator of width K has width KS , with the stated selector price. None of these executable claims uses the latent-preserving maximal coupling. That coupling belongs to the proof of conditional-process convergence, not to the finite-state machine.

These results provide a direct changing-trajectory, likelihood, Girsanov and bounded entropic-value input for a conditional-law theory. They do not assert the strict-dual, rigidity, covariant-form or full kinetic conclusions of a larger program without their separate analytic proofs.

Example 11.5 (Why an amplitude bound is not a tightness argument). Let ξ_1, ξ_2 be independent symmetric signs, and, for $n \geq 5$, let $\mathcal{G}_t^n$ reveal ξ_1 at $1/2 - 2/n$ and ξ_2 at $1/2 - 1/n$. On $[0, 1]$ the bounded martingale

$$M_t^n = \mathbb{E}[\xi_1 + \xi_2 \mid \mathcal{G}_t^n]$$

converges at every finite collection of fixed times to the martingale $M_t = \mathbf{1}_{[1/2, 1]}(t)(\xi_1 + \xi_2)$. Nevertheless its laws are not tight in the Skorokhod J_1 topology. On the event $\xi_2 = -\xi_1$, of probability $1/2$, its path has an excursion of height one on an interval of length $1/n$ and is zero outside that interval. The three-time oscillation modulus of such a path is one at every scale greater than $2/n$. This violates the necessary J_1 tightness modulus criterion. All maximal moments are uniformly bounded by 2^p .

This example concerns the inference from finite-dimensional convergence and a maximal amplitude estimate alone. It does not claim that every possible stronger conditional-kernel hypothesis fails. The uniform estimates in Theorem 11.1 remove the need for that inference in the observation class treated here.

12. GLOBAL TRANSPORT OF THE BACKWARD EQUATION

We retain exactly the common-preparation, bounded-signal and positive-noise hypotheses of Section 11. In particular the canonical joint laws P_n, P are equivalent and have the same latent marginal ν . All comparisons on a single observation path use common measurable versions of the

conditional expectations. The following estimate avoids division by a small observation density in the final bound.

Lemma 12.1 (An integrated posterior comparison). *Let P, Q be equivalent laws on a standard Borel product carrying a latent variable Z and an observation Y , and let $\|P - Q\|_{\text{TV}} \leq \epsilon$. If $|u(Z)| \leq M$ and $\mathcal{F}$ is an observation sub- σ -field, then, under either $S = P$ or $S = Q$,*

$$\mathbb{E}_S[\mathbb{E}_Q[u \mid \mathcal{F}] - \mathbb{E}_P[u \mid \mathcal{F}]] \leq 4M\epsilon, \quad (12.1)$$

$$\mathbb{E}_S[\mathbb{E}_Q[u \mid \mathcal{F}] - \mathbb{E}_P[u \mid \mathcal{F}]]^2 \leq 8M^2\epsilon. \quad (12.2)$$

A common latent marginal is not needed for these two inequalities.

Proof. Use a dominating finite measure on $\mathcal{F}$. Write p, q for the densities of the restricted probability laws and a, b for the densities of the signed measures $\mathbb{E}_P[u\mathbf{1}_A], \mathbb{E}_Q[u\mathbf{1}_A]$ on $\mathcal{F}$. Their conditional means are $m_P = a/p$ and $m_Q = b/q$. Outside null sets, $|m_P|, |m_Q| \leq M$, and

$$p(m_Q - m_P) = m_Q(p - q) + (b - a).$$

The integral of $|p - q|$ is at most 2ϵ , by contraction of total variation. The integral of $|b - a|$ is at most $2M\epsilon$, since multiplication by u followed by restriction cannot increase signed variation by more than M . Integrating the displayed identity proves (12.1) under P . Interchanging the two laws proves it under Q . The squared bound follows from $|m_Q - m_P| \leq 2M$. $\square$

Let $|f| \leq 1$, $\gamma > 0$, and use the notation θ^n, U^n, Z^n, I^n of Proposition 11.2 and Theorem 11.3. Here Z^n denotes the scalar backward integrand, whereas the unadorned latent argument of H_n is the preparation variable. Define the deterministic bounds

$$A_n = \sigma^{-2}\{2\eta_n^2 + 16M^2T\epsilon_n\}, \quad (12.3)$$

$$C_n = \frac{2(1 + e^{4\gamma})}{\gamma^2} A_n, \quad (12.4)$$

where $\epsilon_n = \min\{1, \eta_n/(2\sigma)\}$ as before.

Theorem 12.2 (Global innovation and entropic-integrand transport). *Under the hypotheses of Theorem 11.1, under either physical law $S = P$ or $S = P_n$ on the same canonical path,*

$$\mathbb{E}_S \int_0^T |\theta_t^n - \theta_t|^2 dt \leq A_n, \quad \mathbb{E}_S \int_0^T |Z_t^n - Z_t|^2 dt \leq C_n. \quad (12.5)$$

No stopping at a density level is required. If $\eta_n \rightarrow 0$, both integrals also converge to zero in expectation under the reference law R .

On the single latent-preserving coupling of Theorem 11.1, evaluate each process on its own observation path, denoting these evaluations by a bar. Then

$$\mathbb{E} \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt \leq A_n + 4M^2T\epsilon_n/\sigma^2, \quad (12.6)$$

$$\mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt \leq C_n + 16M^2T\epsilon_n/(\gamma^2\sigma^2). \quad (12.7)$$

If $\eta_n \rightarrow 0$, then $\bar{I}^n \rightarrow \bar{I}$ uniformly in probability. Writing $\bar{J}_t^n = \int_0^t \bar{Z}_s^n d\bar{I}_s^n$ for the stochastic integral in its own observation filtration, and $\bar{J}$ for its limit, one has

$$\mathbb{E} \sup_{t \leq T} |\bar{J}_t^n - \bar{J}_t|^2 \leq 2\mathbb{E} \sup_{t \leq T} |\bar{U}_t^n - \bar{U}_t|^2 + \frac{8M^2T}{\sigma^2} \mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt. \quad (12.8)$$

The first term on the right has the explicit bound in (11.11). Thus the full bounded entropic backward representation, including its martingale integral, is transported globally. The coupling

need not make either innovation Brownian in the joined filtration; each integral is defined under its own marginal law.

Proof. For each deterministic t , under P split

$$\sigma(\theta_t^n - \theta_t) = \mathbb{E}_{P_n}[H_n(t, Z) \mid \mathcal{F}_t] - \mathbb{E}_P[H_n(t, Z) \mid \mathcal{F}_t] + \mathbb{E}_P[H_n(t, Z) - H(t, Z) \mid \mathcal{F}_t].$$

Lemma 12.1, conditional Jensen and $(x + y)^2 \leq 2x^2 + 2y^2$ bound its expected square by $16M^2\varepsilon_n + 2\mathbb{E}_\nu|H_n(t, Z) - H(t, Z)|^2$. Integration gives the first bound in (12.5) under P . Under P_n use the same decomposition with the signal-difference term conditioned under P_n and the law-change term applied to H . The common latent marginal gives the identical bound. Jointly measurable conditional versions, and predictable versions equal for dt -almost every time, give the same integrated statements.

Put $g = e^{\gamma f}$ and $c = \mathbb{E}_\nu g$. Tilt the two full joint laws by $g(Z)/c$, writing P^g, P_n^g . They still have the same latent marginal $\nu_g = g\nu/c$, and their observation signals remain H, H_n . Since $e^{-\gamma} \leq g, c \leq e^\gamma$,

$$\|P_n^g - P^g\|_{\text{TV}} \leq e^{2\gamma}\varepsilon_n, \quad \eta_{n,g}^2 \leq e^{2\gamma}\eta_n^2.$$

Moreover, the reference martingale coefficient $a^{n,g}$ from the proof of Theorem 11.3 satisfies

$$\alpha_t^n := \frac{a_t^{n,g}}{N_t^{n,g}} = \sigma^{-1} \mathbb{E}_{P_n^g}[H_n(t, Z) \mid \mathcal{F}_t].$$

Apply the preceding innovation argument to these tilted laws. It bounds the squared integral of $\alpha^n - \alpha$ by $e^{2\gamma}A_n$ under either tilted physical law. For an observation-measurable nonnegative random variable, passage back to the corresponding original physical law costs at most $e^{2\gamma}$: the full density c/g is bounded by this constant, and so is its observation projection. Consequently

$$\mathbb{E}_S \int_0^T |\alpha_t^n - \alpha_t|^2 dt \leq e^{4\gamma}A_n.$$

The identity $Z^n = (\alpha^n - \theta^n)/\gamma$ proves the second bound in (12.5).

The pointwise bounds, valid for all models, are

$$|\theta^n|, |\alpha^n| \leq M/\sigma, \quad |Z^n| \leq 2M/(\gamma\sigma). \quad (12.9)$$

The squared time integrals are thus uniformly bounded random variables. Convergence in P -probability, followed by absolute continuity of the fixed R with respect to P , gives convergence in R -probability. Boundedness then gives convergence of their R -expectations. This argument establishes global reference convergence without claiming a new explicit reference rate.

For the coupling, let $E_n = \{Y^n = Y\}$, so that $\mathbb{P}(E_n^c) \leq \varepsilon_n$. On E_n both processes are functions of the same observation path. The restricted expectation of their squared difference is at most the corresponding unrestricted expectation under the limiting marginal P . On E_n^c apply (12.9) and integrate for time T . This proves (12.6)–(12.7).

On E_n , the Brownian reference coordinates $Y^n/\sigma, Y/\sigma$ agree, so

$$\sup_{t \leq T} |\bar{I}_t^n - \bar{I}_t|^2 \leq T \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt.$$

Markov's inequality and $\mathbb{P}(E_n^c) \rightarrow 0$ prove uniform convergence in probability. This assertion does not require any moment estimate on the observation difference on the exceptional event.

Finally, the backward equation already proved gives, under each marginal,

$$\bar{J}_t^n = \bar{U}_t^n - U_0 + \frac{\gamma}{2} \int_0^t |\bar{Z}_s^n|^2 ds.$$

The common initial value is $U_0 = \gamma^{-1} \log \mathbb{E}_\nu g$. By (12.9),

$$||\bar{Z}_s^n|^2 - |\bar{Z}_s|^2| \leq \frac{4M}{\gamma\sigma} |\bar{Z}_s^n - \bar{Z}_s|.$$

Subtract the two integral identities, use Cauchy–Schwarz in time and $(x + y)^2 \leq 2x^2 + 2y^2$, and take expectations. This gives (12.8). The integral identity also defines compatible measurable versions on the coupling, without asserting a martingale property for its joint filtration. $\square$

The theorem strengthens the localized conclusion of Theorem 11.3 without changing its terminal condition or its driver. The terminal condition remains the observation-measurable conditional certainty equivalent, not the unobserved latent payoff. Positive fixed noise, bounded signals, and a common preparation are used in the proof. A singular zero-noise limit, an unbounded terminal condition, or a different nonlinear driver would require additional arguments.

13. A UNIFORM CERTIFICATE THROUGH A MOVING COLLISION

We now use all three parts of the argument on one family. The finite marked reference law and the uniform scattering geometry are those of Theorem 9.1 and Lemma 9.2. They are proved above, including the no-collision control. We use a continuous observation of the same detector so that finite reports and conditional processes are generated by one experiment.

Let z denote a preparation in the boxes of (9.3), and let ν be that prepared law. The sign of the impact orientation is $2B - 1$, and the conserved center-of-mass longitudinal sign is the actual mark W . Their correlation is prepared: $\mathbb{P}(B = b, W = w) = 5/16$ for $b = w$ and $3/16$ otherwise. The contact diameter d ranges over $I = [9/10, 11/10]$. Let $X_d(t, z)$ be the two-particle phase path, with outgoing velocities used at the collision, and define

$$H_d(t, z) = h(v_{1,y}^d(t, z)), \quad Y_t^d = \int_0^t H_d(s, z) ds + \sigma B_t^d, \quad 0 < \sigma \leq 1/24.$$

The Brownian noise is independent of the preparation. The source experiment has the same physical preparation and mark but constant reports. The target reports at times 2 and 3 are

$$Z_1^d = \mathbf{1}\{2(Y_2^d - Y_{3/2}^d) > 0\}, \quad Z_2^d = A\mathbf{1}\{2(Y_3^d - Y_{5/2}^d) > 0\}. \quad (13.1)$$

The acquisition gate $A \in \{0, 1\}$ is chosen after Z_1^d ; it does not act on the mechanical trajectory. The continuous path is the auxiliary observation experiment for conditional consumers; the finite simulator is not given this path or an uncharged buffer for computing its increments. The target sensor implements the declared window acquisition.

Lemma 13.1 (Quantitative change of the microscopic path). *For all preparations in the specified boxes, the collision time $t_c(d)$ and outgoing tagged velocities obey*

$$|t_c(d') - t_c(d)| \leq |d' - d|, \quad |v_i^+(d') - v_i^+(d)| \leq 30|d' - d| \quad (i = 1, 2).$$

Writing $\Delta = |d' - d|$, one has

$$\sup_z \int_0^3 |H_{d'}(t, z) - H_d(t, z)|^2 dt \leq \Delta + 2700\Delta^2. \quad (13.2)$$

The phase paths converge uniformly over the preparation in the Skorokhod J_1 metric as $d' \rightarrow d$. More precisely, for the product metric taking the maximum of the per-particle Euclidean position and velocity distances, a time change aligning the collision times gives J_1 distance at most 100Δ .

Proof. Write r and g for the incoming relative position and velocity, and $n(d) = (r + t_c(d)g)/d$ for the contact normal. The collision root satisfies $|r + t_c g| = d$. The nongrazing estimate from Lemma 9.2 is $-g \cdot n > 1443/1000$. Differentiating the root with respect to d gives

$$t'_c(d) = \frac{1}{g \cdot n(d)}, \quad |t'_c(d)| < 1.$$

The incoming bound is $|g| < 3$, so

$$|n'(d)| \leq \frac{|t'_c(d)| |g| + 1}{d} < 5.$$

Since $v_1^+ = V + g/2 - (g \cdot n)n$ and $v_2^+ = V - g/2 + (g \cdot n)n$, differentiation bounds either velocity derivative by $2|g| |n'| < 30$.

Before collision the detector is zero, and after collision it is a constant of absolute value at most one. Its Lipschitz constant is one. The two signals therefore differ on the interval between the two collision times by at most one and afterwards by at most 30Δ . The intervening interval has length at most Δ , and the whole observation lasts three units. This proves (13.2).

Choose the increasing piecewise affine time change fixing 0, 3 and sending $t_c(d)$ to $t_c(d')$. Its uniform displacement is at most Δ . Aligned velocities differ by at most 30Δ , with the same outgoing convention at the jump. Each particle's speed is below four and its velocity jump has norm below three. Without time change, the position difference is bounded by $3\Delta + 3(30\Delta) = 93\Delta$, by integrating the velocity difference before and after the two jump times. Aligning time adds at most 4Δ . The asserted bound follows, with a harmless rounding to 100Δ . $\square$

Theorem 13.2 (Microscopic-to-audit and conditional-law composition). *For the single family (13.1), all of the following hold.*

(a) *At the same width-one private register,*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \quad (r \geq 1), \quad \delta_2^h < 13/100,$$

uniformly for $d \in I$ and $0 < \sigma \leq 1/24$. When $d \leq 2/5$ the detector is zero on $[0, 3]$, and all three finite-experiment deficiencies are zero.

(b) *There is one explicit rational reset-audit rule with one all-on training row, eight marked outcomes and $n = 2800$ training runs, whose expected validation score exceeds $2/5$ against every private width-one simulator throughout that same parameter region. In detail, for counts C choose an event $A_C \subseteq \{0, 1\}^3$ maximizing*

$$Q_*(A_C) - C(A_C)/2800, \tag{13.3}$$

with lexicographic tie breaking. Here Q_ is the rational marked reference law of (B, B, W) . The independent validation uses the actual physical target, not Q_* . This finite rule has at most $\binom{2807}{7}$ training-count states at any scheduled epoch; a conservative complete auditor bound is $8(256 + 1)\binom{2807}{7}$. The state cost belongs to the auditor. The certificate is an expected-score inequality, not a confidence claim about one realized score.*

(c) *Fix $\sigma > 0$ and let $d_n, d \in I$ with $d_n \rightarrow d$. Set*

$$\epsilon_n = \min \left\{ 1, \frac{\sqrt{|d_n - d| + 2700|d_n - d|^2}}{2\sigma} \right\}. \tag{13.4}$$

The joint preparation–continuous-observation laws differ by at most ϵ_n . Every finite resource deficiency above changes by at most ϵ_n , and a fixed audit changes by at most ϵ_n because its source behavior is unchanged. On a latent-preserving coupling, $X_{d_n} \rightarrow X_d$ in J_1 and each bounded posterior converges in supremum L^2 with bound $84\epsilon_n$. The full and projected likelihoods converge in reference supremum L^2 , and jointly in probability with the microscopic paths on that coupling.

The projected stochastic exponentials, bounded entropic backward values, localized integrands and stopping consumers have the conclusions of Proposition 11.2, Theorem 11.3, and Corollary 11.4.

(d) *On compact parameter sets with $\sigma \geq \sigma_0 > 0$, the finite marked probabilities have uniform rational enclosures. The observable chart, strict same-budget polynomial verification and rational upper tables are therefore effective in the exact sense of Propositions 8.1 and 7.6.*

Proof. Every collision occurs before time $3/2$. Conditional on preparation, the two normalized window increments in (13.1) are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. The outgoing detector has sign $2B - 1$ and absolute value greater than $1/4$. Coupling each noisy report to B gives total marked variation less than $1/300$ from the exact finite marked experiment, uniformly for every acquisition gate. The finite calculation of Theorem 9.1 gives

$$d_* = \sqrt{5/2} - 9/8, \quad \delta_*^p = \delta_*^v = d_*, \quad \delta_{*,2}^h = 1/8.$$

Thus private and visible values exceed $d_* - 1/300 > 9/20$, and the hidden value is less than $1/8 + 1/300 < 13/100$. The two-valued hidden witness stores its selector, not an observed mark. In the no-collision regime both window reports before gating are independent fair noise, independent of W . Fresh private coins reproduce them exactly, including feedback gates. This proves (a).

For (b), use only the all-on feedback row. Any private width-one simulator in this row emits independent bits with probabilities p, q , independently of the actual W . The calculation in Theorem 9.1 already gives minimum variation d_* for this row. Its alphabet (z_1, z_2, W) has eight elements, hence $D = 7$. The explicit empirical audit in Theorem 4.1, trained using Q_* , has expected ideal score at least

$$d_* - \sqrt{7/2800} = d_* - 1/20.$$

The candidate training and validation laws do not change when the target is replaced by the physical target. Conditional on any selected event, its target probability changes by less than $1/300$. Therefore the actual score is strictly greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (13.5)$$

For completeness, the last strict inequality is the positive-root comparison $\sqrt{5/2} > 947/600$; its squared margin is $5/2 - (947/600)^2 = 3191/360000 > 0$. All probabilities in Q_* are 0, $3/16$ or $5/16$, so (13.3) is a finite rational rule independent of d, σ, p, q . At training epoch k the number of eight-category count vectors is $\binom{k+7}{7} \leq \binom{2807}{7}$. A buffer of eight states records the current two reports and terminal mark; 256 event masks and an idle symbol suffice during event selection and validation. The conservative product bound follows. The clock and the source-reset interface are the declared resources. In particular the proof does not enumerate or claim to store a small complete coefficient table.

For (c), Lemma 13.1 gives the signal-distance hypothesis of Theorem 11.1, with $T = 3$ and $M = 1$. It also supplies its microscopic J_1 hypothesis. That theorem gives (13.4) and all the joint process assertions. The finite acquisition is a common measurable causal map of the continuous observation, so marked variation contracts for every gate. The source is blind and independent of d . Consequently its executable candidate behaviors are identical across d , and Proposition 4.4 has $\alpha = 0$ and $\beta = \epsilon_n$. The same same-budget comparison applies in each visible fiber and in each hidden mixture. The exponential and backward claims now follow from the identified projected density and the explicitly bounded entropic consumer. This is where the finite certificate and the conditional-law theorem meet the same microscopic approximation.

Finally, the collision root is uniformly separated from grazing, its formula is algebraic on the rational preparation boxes, and the normalized window noise has variance $2\sigma^2$ bounded away from zero on the stated compact sets. The box subdivision and Gaussian-tail enclosure argument

in Proposition 9.4 therefore applies with that variance. It returns rational enclosures with a computable uniform error. The rational quotient algorithm and same-budget strict certificate procedure then prove (d). Uniformity as $\sigma \downarrow 0$ is not asserted. $\square$

The all-on audit in part (b) is a single witness for the lower bound in the full feedback problem: a maximum over feedback trees includes this row. The visible lower bound is instead the exact fiber theorem; no claim is made that an auditor with a seed-independent state bound can store an arbitrarily large visible-seed transcript. This separates the two resource budgets rather than silently moving cost from one to the other.

The finite certificate is also preserved under an additional rational presentation error ρ : its lower score decreases by at most ρ when only the target is changed, and by the reset-cost estimate when the source is changed. Thus the certificate is an effective consumer of the physical presentation theorem, not an isolated exact finite example.

Theorem 13.2 and the preceding results prove Theorem A.1. The C2-type consequence is a joint changing-trajectory/likelihood/posterior theorem with an identified stochastic exponential and a fully specified bounded entropic backward consumer. Nothing in that conclusion identifies geometric similarity with kinetic scaling, or supplies particle-uniform recollision estimates by renaming this fixed-particle model.

14. A FINITE-REGISTER CERTIFICATE TRANSPORTED THROUGH THE MICROSCOPIC LIMIT

The general empirical audit in Theorem 4.1 treats every coordinate of a marked row as unknown. The private continuation constraint in the collision experiment forces a smaller family. Exploiting this constraint yields a much smaller executable auditor, without granting it knowledge of the candidate law.

In the all-on row of Section 13, every private width-one candidate emits two independent bits with parameters p, q , independently of the fair actual mark W . Put

$$a = (1 - p)(1 - q), \quad c = pq.$$

The ideal target Q_* is supported on 000, 001, 110, 111, with respective masses $5/16, 3/16, 3/16, 5/16$. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. Every other atom has target mass zero and cannot improve the maximizing event for $Q_* - b$.

Lemma 14.1 (Two-count audit and its exact register implementation). *In $n \geq 1$ independent all-on candidate trials let C_0, C_1 count report pairs 00, 11, respectively, ignoring the training mark. On the four supported atoms use the empirical masses $C_0/(2n), C_0/(2n), C_1/(2n), C_1/(2n)$, and select exactly those atoms whose target mass is strictly greater than this empirical mass. Call the resulting event A_C . For every $p, q \in [0, 1]$,*

$$\mathbb{E}\{Q_*(A_C) - b(A_C)\} \geq \|Q_* - b\|_{TV} - n^{-1/2}. \quad (14.1)$$

For $n = 400$, the rule is implemented by the two truncated counts $\min(C_0, 250), \min(C_1, 250)$, with comparisons to 150 and 250. Including execution of complete transcripts and the two independent validation trials, it has a deterministic streaming realization with at most 504008 retained states at any scheduled epoch. Thus a nineteen-bit register suffices. The schedule clock is free; no other history store is free. The table description and transition evaluation cost are not asserted to be bounded by nineteen bits.

Proof. On any supported atom let $x = Q_*(w) - b(w)$ and let $\hat{x}$ be the empirical discrepancy. If the sign rule selects a different event from an exact maximizing event, the lost contribution is at most $|x - \hat{x}|$. On unsupported atoms both rules can omit the atom. Summing gives regret at most

$$|C_0/n - a| + |C_1/n - c|.$$

The two counts need not be independent. Their marginal binomial variances and Cauchy–Schwarz give expected regret at most

$$\frac{\sqrt{a(1-a)} + \sqrt{c(1-c)}}{\sqrt{n}} \leq \frac{1}{\sqrt{n}}.$$

This proves (14.1).

For the 00 pair, select both marked atoms when $C_0 < 150$, only the atom of target mass 5/16 when $150 \leq C_0 < 250$, and neither when $C_0 \geq 250$. The same rule applies to 11, with the other mark being the atom of larger target mass. At equality the zero empirical discrepancy is excluded. Increasing a count beyond 250 cannot change the final choice. Two registers with 251 possibilities each therefore implement the rule, with at most nine possible final event masks.

A buffer with eight possibilities is sufficient to execute and record a completed two-bit-plus-mark transcript. During training the pair of truncated counts and this buffer use at most $8 \cdot 251^2 = 504008$ states. After training, discard the counts and retain the event mask. During validation at most nine masks, one target-event indicator with two values, and the same eight-state transcript buffer are needed, a total of 144 states. The final signed score takes three values. Since the epoch is known, labels can be reused between these phases; their state counts need not be multiplied. The update maps and threshold comparisons are parameter-independent deterministic maps. They are thus stochastic rows with entries zero or one, or equivalently a compatible continuation cover by Theorem 3.2. Fresh sampling randomness belongs to the specified reset experiments, not to an uncharged auditor register. Finally $504008 < 2^{19}$. $\square$

Theorem 14.2 (Finite-register microscopic certification and global transport). *Consider the marked hard-sphere family of Theorem 13.2, with $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$. There is one rational streaming audit, independent of d, σ, p, q , using 400 candidate training resets and one independent candidate and target validation trial, whose expected signed score is strictly greater than $2/5$ against every private width-one candidate. It uses at most 504008 retained auditor states. The candidate remains private width one; the auditor’s register is a distinct, charged resource. The physical deficiencies still satisfy*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \ (r \geq 1), \quad \delta_2^h < 13/100.$$

For $d \leq 2/5$ all three finite-experiment deficiencies vanish.

Fix a positive σ and a sequence $d_k \rightarrow d$ within the collision interval. Put

$$\rho_k = |d_k - d| + 2700|d_k - d|^2, \quad \epsilon_k = \min\{1, \sqrt{\rho_k}/(2\sigma)\}.$$

The same 400-reset rule changes its expected score by at most ϵ_k under this perturbation. At the same time the microscopic paths, full and projected likelihoods, posteriors, and bounded entropic values have the joint convergence of Theorem 13.2. Their innovations, backward integrands and backward martingale integrals additionally have all the global conclusions of Theorem 12.2, with $T = 3$, $M = 1$, $\eta_k^2 \leq \rho_k$, and $\epsilon_k \leq \epsilon_k$. On compact parameter sets bounded away from zero noise, the same audit has the effective probability-enclosure and rational-verification properties of Theorem 13.2(d).

Proof. The private width-one continuation restriction gives precisely the candidate family used in Lemma 14.1; the actual mark is not available to its transition rows. The exact finite separation in Theorem 9.1 gives

$$\min_{p,q} \|Q_* - b(p, q)\|_{\text{TV}} = d_* := \sqrt{5/2} - 9/8.$$

The scattering and observation estimates proved in Theorem 13.2 put the physical marked target within $1/300$ of Q_* , uniformly in all stated parameters. Train the two-count rule against the

fixed rational Q_* , but use the actual physical target for validation. Each selected event's target probability changes by less than $1/300$. Hence its actual expected score is greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (14.2)$$

The last strict inequality is exact: it is equivalent to $\sqrt{5/2} > 947/600$, whose squared margin is $3191/360000$. The rule's executable memory bound was proved in Lemma 14.1. Its total trial count is 402, not 400 once validation is included. The private, hidden and visible deficiency bounds and the no-collision control are inherited from the same physical theorem, rather than inferred from an auditor whose signature was changed.

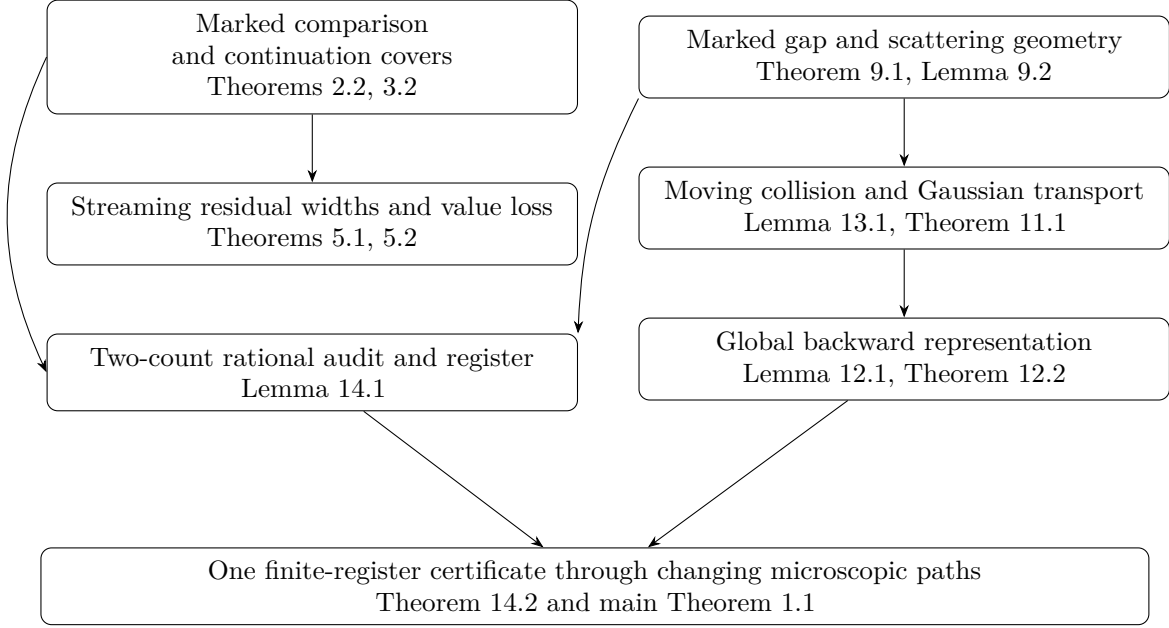
For changing d_k , Lemma 13.1 supplies the signal modulus ρ_k and the pathwise J_1 bound $100|d_k - d|$. Theorem 11.1 gives the joint marked variation bound ϵ_k and the likelihood/posterior convergence. The candidate source is blind and its behavior does not depend on d . Thus training laws and candidate validation laws are unchanged, and Proposition 4.4 has $\alpha = 0$ and $\beta = \epsilon_k$. There is no factor of 400 in this transport cost. The same preparation and signal bounds verify every hypothesis of Theorem 12.2, proving the global backward assertions. Finally the audit uses rational thresholds and finitely many deterministic row maps. The positive-noise enclosure theorem therefore applies directly; an additional target-presentation error ζ reduces its score by at most ζ . $\square$

The two-count construction is specific to the private behavior family. A hidden mixture can destroy the independence of its two reports, and its mark relation must still be checked. The bound is therefore not asserted against the hidden class. The visible lower bound has its separate fiberwise proof; an arbitrarily large visible alphabet is not silently stored in the nineteen-bit auditor. The certificate concerns expected score, not a confidence statement about a single realized validation pair. Repeating validation for statistical confidence would incur an additional sample and storage budget.

The theorem uses a single mechanism: the continuation restriction fixes the feasible candidate laws and permits the compressed audit; scattering supplies its marked separation; common-preparation variation transports the same executable test and the global conditional-law consumer through a change in the microscopic trajectory. The collision transduces a prepared impact variable into a signal. It does not create the prepared correlation between that variable and the mark, and it is not a particle-number limit.

APPENDIX A. THE CANONICAL PROOF CHAIN

The diagram separates exact realization and testing complexity from the physical verification of the common-preparation hypotheses. Both feed the same final certification theorem. A dependency arrow means that the stated proof uses the source result, not that the source is the only possible method of proof.



The binary optimal-value theorem is the classification and lower-bound branch of the main result; it is not needed to prove the two-count upper bound. The latter uses the deterministic continuation realization and the specific private behavior geometry directly. The finite-reset minimax and its moment support bound describe the more general audit framework. The exact nonlinear witness, revealed-behavior hierarchy and rational chart retain their separate information conventions and full proofs.

The comparison theorem, positive continuation realization, erasure-cut law, reset minimax and sampling obstruction, chart, nonlinear witness, marked scattering gap, posterior estimate, moving-collision estimate and localized entropic theorem are inherited inputs. The stochastic residual error inequality, exact binary audit-width threshold, compressed certificate and global backward-integrand bounds are the additions of this revision. The preceding introduction states which proof mechanisms are classical. None of the label, source, build or diagnostic checks accompanying this article is an independent verification of the analytic proofs.

The actual mark is never exposed to the online simulator. The reset auditor has a separate completed-trial interface. A revealed-behavior test sees an exact law. The common-latent maximal coupling may use an entire observation path and is not an executable causal simulator. These privileges are not interchangeable.

A.1. Inherited baseline. The following summary is retained from the predecessor. The present article strengthens its memory and backward-integrand conclusions in Theorem 1.1; the earlier 2800-trial audit remains valid.

Theorem A.1 (Finite-resource certification and microscopic transport). *The following assertions hold with the conventions just described.*

(i) *Finite causal kernels have a register realization with a prescribed width profile exactly when their normalized continuation laws admit compatible finitely generated polytopes of those widths. At an erasure cut carrying a latent letter of law π , the optimal error with K retained states is*

$$1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}, \quad \pi_{(1)} \geq \cdots \geq \pi_{(N)}.$$

Independent hidden or visible selectors do not improve this cut value.

(ii) *The reset-audit value is attained, has a finite-dimensional moment minimax representation on the original B , and satisfies*

$$0 \leq \delta(B, Q) - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}, \quad D = \sum_j (|\Omega_j| - 1).$$

The exponent $1/2$ cannot be improved uniformly over compact candidate classes. This assertion is about reset samples, not about the minimal degree of unrestricted polynomial tests.

(iii) *Suppose two prepared microscopic observation models have a common preparation variable, bounded signals H, H' on $[0, T]$, and the same positive Gaussian noise level σ . Set*

$$\eta^2 = \mathbb{E} \int_0^T |H - H'|^2 dt, \quad \varepsilon = \min\{1, \eta/(2\sigma)\}.$$

Their joint marked observation laws differ by at most ε . This estimate transports resource spectra at their stated budgets and transports each fixed finite audit with its explicitly charged reset cost. It also gives whole-process posterior convergence. The unprojected and projected likelihoods converge in the supremum L^2 norm under the common Wiener reference when $\eta \rightarrow 0$. For bounded latent payoffs, the associated entropic backward equations converge in value, and their integrands converge after the stated denominator localization.

(iv) *In the two-sphere model of Theorem 13.2, one rationally specified audit with 2,800 training runs has expected discrepancy strictly greater than $2/5$ against every private width-one simulator, uniformly for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. A hidden two-valued selector has error less than $13/100$; every visible-selector width-one error exceeds $9/20$. For $d_n \rightarrow d$ the microscopic paths, likelihoods and posterior processes converge in the precise senses of that theorem, and the same audit remains valid. For $d \leq 2/5$ all three deficiencies vanish.*

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [3] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.
- [4] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [5] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.
- [6] I. Gallagher, L. Saint-Raymond and B. Texier, *From Newton to Boltzmann: Hard Spheres and Short-Range Potentials*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, 2014. The flow and boundary discussion used here is in Sections 4.1–4.3 of arXiv:1208.5753v2.
- [7] C. Widder, J. Zimmer and T. Schilling, On the generalized Langevin equation and the Mori projection operator technique, preprint (2025), revised 2026. arXiv:2503.20457v6.
- [8] C. Widder and T. Schilling, Generalised Langevin dynamics: significance and limitations of the projection operator formalism, preprint (2026). arXiv:2604.20453v2.
- [9] H. F. Trotter, Approximation of semi-groups of operators, *Pacific J. Math.* **8** (1958), 887–919. doi:10.2140/pjm.1958.8.887. Theorem 5.2, p. 902, is the approximation theorem cited here.
- [10] H. Weisshaupt, LeCam’s randomization criterion in the setting of locally convex spaces without lattice structure, EURANDOM report 2002–017, 22 April 2002. Lemma 1 and Theorem 5, pp. 14–16. Author report.
- [11] S. Basu, Algorithms in real algebraic geometry: a survey, author preprint, arXiv:1409.1534 (2014). Published in *Panoramas et Synthèses* **51** (2017), 107–153. Theorem 2.1 and Section 2.1 in the author preprint are used here.
- [12] M. C. Paull and S. H. Unger, Minimizing the number of states in incompletely specified sequential switching functions, *IRE Trans. Electron. Comput.* **EC-8** (1959), 356–367. Publisher record.
- [13] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.

- [14] N. Saldi, A topology for team policies and existence of optimal team policies in stochastic team theory, preprint (2017), arXiv:1711.00639.
- [15] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures and optimal solutions, author preprint arXiv:1609.07685v2.
- [16] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.
- [17] S. Kiefer, Notes on equivalence and minimization of weighted automata, preprint (2020), arXiv:2009.01217v1.
- [18] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176. doi:10.2140/pjm.1958.8.171.
- [19] H. Weisshaupt, A generalization of Le Cam’s randomization criterion, *Studia Sci. Math. Hungar.* **43** (2006), 443–486. doi:10.1556/SScMath.43.2006.4.3. Theorem numbers refer to the author’s 39-page preprint, Randomization.pdf.
- [20] E. de Klerk and M. Laurent, Error bounds for some semidefinite programming approaches to polynomial minimization on the hypercube, *SIAM J. Optim.* **20** (2010), 3104–3120. doi:10.1137/100790835.
- [21] M. Laurent and L. Slot, An effective version of Schmüdgen’s Positivstellensatz for the hypercube, arXiv:2109.09528v2, 2023.
- [22] E. de Klerk and J. C. Vera, The link between 1-norm approximation and effective Positivstellensätze for the hypercube, *Numer. Algebra Control Optim.* **16** (2026), 84–104. doi:10.3934/naco.2024060. Theorem numbers refer to arXiv:2404.04190v2.
- [23] M. Fliess, Séries rationnelles positives et processus stochastiques, *Ann. Inst. H. Poincaré Sect. B* **11** (1975), 1–21. Original article. Theorems 3.1 and 5.4 and Corollary 2, pp. 8–9 and 18–19.